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Risk Prediction in Chronic Kidney Disease[a]

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CHAPTER OUTLINE

KEY POINTS

- Because of a large variance in chronic kidney disease (CKD) progression, there is a need to develop methods for risk stratification within CKD to identify the subgroup of patients who are at risk of progression to end-stage kidney disease and who may benefit the most from intervention.
- The combination of demographic risk factors (age, sex) and biomarkers (laboratory variables including albuminuria and hemoglobin) can be useful in risk stratification of patients with CKD.
- In a similar manner to the Framingham risk score for prediction of cardiovascular risk, kidney risk scores such as the kidney failure risk equation have allowed for the accurate prediction of CKD progression.
- Equations such as the kidney failure risk equation are externally validated and ready to be implemented in a clinical setting.

Chronic kidney disease (CKD) currently affects more than 30 million Americans[1] and 840 million people globally. A meta-analysis of studies from around the globe has reported a prevalence for CKD stages 1–5 among adults of 8.7% to 18.4% with a global mean prevalence of 13.4% (95% confidence interval [CI], 11.7% to 15.1%).[2] The large number of people known to be affected by CKD has major implications for the provision of health care—in particular, nephrology services. In the past 2 decades, nephrology has moved from a position where it provided highly specialized services to a relatively small number of patients with specific and relatively rare kidney disease or advanced CKD, to one where it must concern itself with the care of less advanced CKD in a substantial proportion of the general population. Furthermore, early stage CKD is largely asymptomatic, and detection therefore requires a screening process. Previous studies have indicated that screening whole populations is not cost-effective[3] except in higher-income countries,[4] and a means of identifying high-risk subgroups for targeted screening is therefore required. Successful screening programs are likely to identify large numbers of patients with previously undiagnosed CKD; however, in most countries, nephrology services are unable to provide long-term care to all persons with CKD, and the associated costs would be prohibitive. A solution to

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this problem was suggested by studies showing that there is substantial heterogeneity among patients who meet the diagnostic criteria for CKD, with most being at relatively low risk of ever progressing to end-stage kidney disease (ESKD). One reason for this heterogeneity is the use of an estimated glomerular filtration rate (eGFR) <60 mL/min/1.73 m^2 as the standard of CKD diagnosis in clinical practice.[5] This provides susceptibility to both overdiagnosis and underdiagnosis of CKD as any given level of eGFR will have a large variation in the risk for progression to kidney failure.[6] As a result, there will be situations in which low-risk patients are overrecognized and receive unnecessary treatment or, conversely, high-risk patients being under-recognized and receiving less-than-optimal treatment with regard to CKD progression.[7] The Kidney Disease Outcomes and Quality Initiative (K/DOQI) classification system for CKD was widely adopted and proved valuable, particularly for identifying the prevalence of different stages of CKD in epidemiologic studies.[8] It was noted, however, that the classification provided limited information on the future risk of decline in kidney function.[9] The Kidney Disease Improving Global Outcomes (KDIGO) classification system therefore modified the K/DOQI system so that categories defined by eGFR and albuminuria do correlate with risk,[10] but this does not provide accurate, individual risk prediction. Previous studies identified a wide range of rates of decline in GFR among patients with CKD, and up to 15% may even show an increase in eGFR over time.[11]

In a retrospective cohort study done by O'Hare and colleagues[12] using patients with advanced CKD who progressed to kidney failure from the U.S. Renal Data System and Veterans Affairs databases, the following trajectories were discovered: 1. consistently low eGFR <30 mL/min/1.73 m^2 with a mean yearly decrease of 7.7 mL/min/1.73 m^2, 2. progressive loss of initial eGFR at stage 3 CKD (eGFR of 30–59 mL/min/1.73 m^2) with a mean yearly decrease of 16.3 mL/min/1.73 m^2, 3. accelerated loss of initial eGFR >60 mL/min/1.73 m^2 with a mean yearly decrease of 32.3 mL/min/1.73 m^2, and 4. dramatic loss of initial eGFR >60 mL/min/1.73 m^2 in 6 months or less, with a mean yearly decrease of 50.7 mL/min/1.73 m^2. In a similar study, Li and colleagues[13] also found distinct patterns of CKD progression, this time using patients from the African American Study of Kidney Disease and Hypertension (AASK), which included 846 patients that were followed over a 3-year period. The patterns of progression included 1. stable or increasing eGFR, defined as <2 mL/min/1.73 m^2 decrease per year, 2. rapid decrease in eGFR, defined as a loss of more than 4 mL/min/1.73 m^2 per year, and 3. alternating combinations of the previously mentioned categories. It was determined that a stable or increasing eGFR was the most common pattern of CKD progression, with 58% of participants following such a trajectory.[13] Due to such a variance in progression of CKD, there is a need to develop methods for risk stratification within CKD to identify the relatively small subgroup of patients who are at risk of progression to ESKD and who may benefit from specialist intervention to slow or halt disease progression. Such risk stratification would be equally important for identifying individuals who are at low risk for progression and who could be reassured and spared unnecessary referral to a nephrologist.

Another important aspect of CKD is its association with a substantially increased risk of future cardiovascular events (CVEs) that in most patients with mild CKD substantially exceeds the risk of ESKD.[14] Whereas CKD is associated with a high prevalence of many traditional risk factors for cardiovascular disease (CVD), such as hypertension and dyslipidemia, risk prediction tools such as the Framingham risk score substantially underestimate cardiovascular risk in patients with CKD.[15] It has been proposed that this observation is due to the role of several nontraditional cardiovascular risk factors that are specific to CKD. The importance of CKD and these risk factors has been recognized in the new AHA-PREVENT CVD risk prediction equations.[16]

From the aforementioned discussion, it is clear that there is a need to identify and understand factors associated with an increased risk of developing CKD and, once diagnosed, factors associated with an increased risk of progression to ESKD and CVE. In this chapter, we review current knowledge of these risk factors and the methods being applied to predict risk in CKD patients. Risk factors for CVD in patients with CKD, many of which overlap with risk factors for CKD progression, are discussed in Chapter 42.

DEFINITION OF A RISK FACTOR

A risk factor is a variable that has a causal association with a disease or disease process such that the presence of said variable in an individual is associated with an increased risk of the disease being present or developing in the future. Thus risk factors are a useful tool for identifying persons with increased risk for a disease or particular outcome due to a disease process. In the course of epidemiologic research, many variables may show associations with a disease of interest, but these may be chance associations, noncausal associations, or causal associations (true risk factors). The Bradford Hill criteria provide minimum requirements to be fulfilled to identify a causal relationship between a putative risk factor (exposure) and a disease (outcome; Table 19.1). In complex diseases such as CKD that result from the combined effects of multiple factors, it is likely that many risk factors will not fulfill all criteria. Nevertheless, they do provide a useful framework for assessing the strength of a proposed causal relationship between risk factor(s) and disease.

RISK FACTORS AND MECHANISMS OF CHRONIC KIDNEY DISEASE PROGRESSION

It has been appreciated for several decades that once GFR has decreased to below a critical level, it tends to progress relentlessly toward ESKD. This observation suggests that loss of a critical number of nephrons provokes a vicious cycle of further nephron loss. Detailed studies have elucidated a number of interrelated mechanisms that together contribute to CKD progression, including glomerular hemodynamic responses to nephron loss (raised glomerular capillary hydraulic pressure and single-nephron GFR [SNGFR]), proteinuria, and proinflammatory responses. A generally good prognosis after unilateral nephrectomy[17] attests to the fact that a single pathogenic factor may be insufficient to initiate progressive CKD, but the multihit hypothesis proposes that multiple factors interact to overcome functional reserve and provoke progressive nephron loss.[18] To meet the Bradford Hill criteria of plausibility and coherence, a putative risk factor should therefore somehow affect known mechanisms of CKD progression (see Chapter 50 for further details). Fig. 19.1 shows how risk factors may interact with pathophysiologic mechanisms to initiate or accelerate CKD progression.

On the basis of our understanding of the mechanisms underlying the pathogenesis of CKD and its progression, risk factors may be divided into susceptibility factors, initiation factors, and progression factors (Table 19.2). First, susceptibility includes risk factors associated with an increased risk of an individual developing CKD after exposure to a factor that has potential to cause kidney damage. An example is a reduced nephron number after uninephrectomy, which is associated with an increased risk of developing diabetic nephropathy if the individual develops diabetes.[19] Initiation factors directly cause or initiate kidney damage in a susceptible individual. Examples include exposure to nephrotoxic drugs, urinary tract obstruction, or primary glomerulopathies that may provoke CKD in some (but not all) exposed individuals. Finally, progression factors are those that contribute to the progression of kidney damage once CKD has developed. An example is hypertension, which exacerbates raised intraglomerular hydraulic pressure and therefore accelerates glomerular damage. Studies investigating progression factors should recruit subjects with relatively early stage CKD in a cohort study design. Nevertheless, distinguishing among these categories may in some cases be difficult because some factors (e.g., diabetes mellitus [DM]) may act in all three ways, and in some studies, it may be impossible to separate susceptibility factors from progression factors due to inadequate characterization of participants at study entry.

Table 19.1 Bradford Hill Criteria of Causality

Parameter	Explanation
Strength of association	The stronger the association, the more likely the relationship is causal.
Consistency	A causal association is consistent when replicated in different populations and studies.
Specificity	A single putative cause produces a single effect.
Temporality	Exposure precedes outcome (i.e., risk factor predates disease).
Biological gradient	Increasing exposure to risk factors increases risk of disease, and reduction in exposure reduces risk.
Plausibility	The observed association is consistent with biologic mechanisms of disease processes.
Coherence	The observed association is compatible with existing theory and knowledge in a given field.
Experimental evidence	The factor under investigation is amenable to modification by an appropriate experimental approach.
Analogy	An established cause-and-effect relationship exists for a similar exposure or disease.

Modified from Hill AB. The environment and disease: association or causation? *Proc R Soc Med*. 1965;58:295–300.

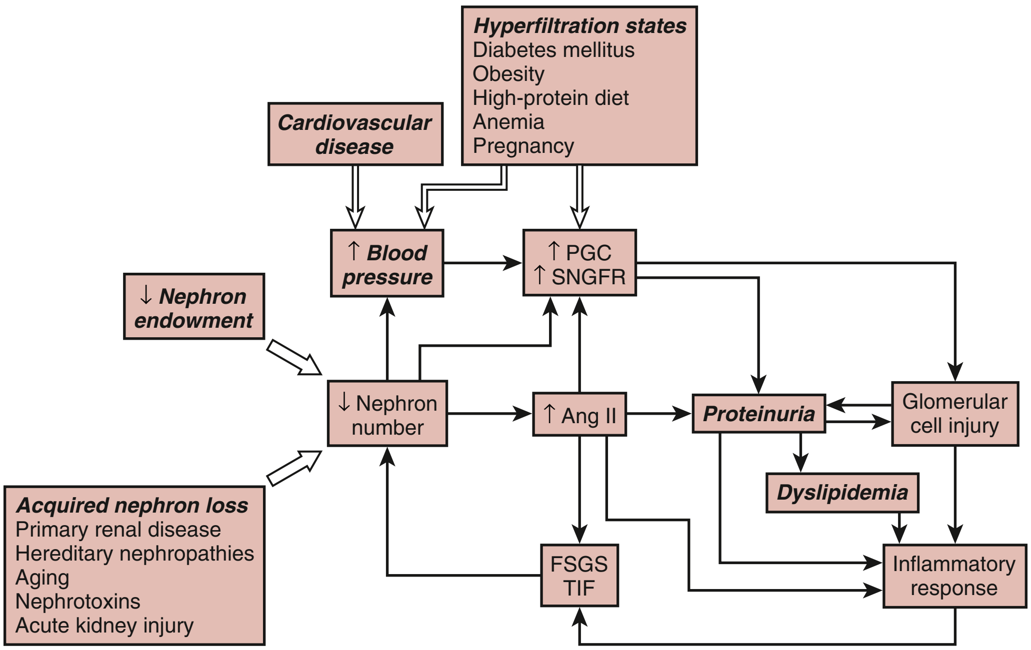

Fig. 19.1 Schematic representation showing the interaction of risk factors for chronic kidney disease progression with pathophysiologic mechanisms that contribute to a vicious cycle of progressive nephron loss. *Ang II,* Angiotensin II; *FSGS,* focal segmental glomerulosclerosis; *PGC,* glomerular capillary hydraulic pressure; *SNGFR,* single-nephron glomerular filtration rate; *TIF,* tubulointerstitial fibrosis. (Adapted from Taal MW, Brenner BM. Predicting initiation and progression of chronic kidney disease: developing renal risk scores. *Kidney Int*. 2006;70:1694–1705.)

Table 19.2 Risk Factors for Chronic Kidney Disease

Risk Factor	Susceptibility	Initiating	Progression
Older age	+		
Sex	+		
Ethnicity	+		+
Family history of chronic kidney disease	+		
Metabolic syndrome	+		
Hemodynamic Factors			
Low nephron number	+		+
Diabetes mellitus	+	+	+
Hypertension	+		+
Obesity	+		+
High protein intake	+		+
Pregnancy		+	+
Primary renal disease		+	
Genetic renal disease		+	
Urologic disorders		+	
Acute kidney injury		+	+
Cardiovascular disease	+		+
Albuminuria			+
Hypoalbuminemia			+
Anemia	+		+
Dyslipidemia	+		+
Hyperuricemia	+		+
↑Asymmetric dimethylarginine			+
Hyperphosphatemia			+
Low serum bicarbonate			+
Smoking	+		+
Nephrotoxins		+	+

DEMOGRAPHIC VARIABLES

AGE

The prevalence of CKD increases with age and is reported to be as high as 56% in those 75 years or older.[20] Longitudinal studies of subjects without kidney disease have observed a decline in GFR with advancing age in some individuals, implying that nephron loss may be regarded as part of normal aging.[21] Additionally, aging is associated with an increase in several other risk factors for CKD including hypertension, obesity, and CVD, which may contribute to the rise in CKD prevalence. Several population-based studies have found a higher incidence of proteinuria and CKD,[22-24] as well as ESKD with older age.[25] Similarly, the incidence of a decline in kidney function over 5 years was greater among older compared with younger patients with hypertension.[26] One study reported that older age is associated with a lower risk of treated kidney failure among patients with CKD, even if they have accelerated decline in eGFR.[27] This apparent contradiction is most likely explained by the competing risks of death and ESKD in older patients, illustrated by the

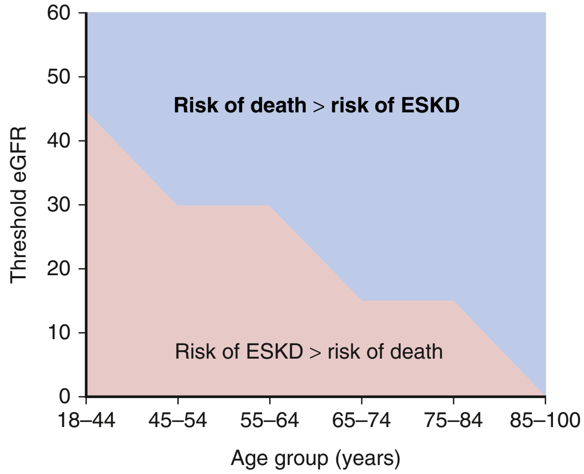

Fig. 19.2 Baseline estimated glomerular filtration rate (eGFR) threshold. Below this, the risk for end-stage kidney disease (ESKD) exceeded the risk for death in each age group among 209,622 U.S. veterans with chronic kidney disease stages 3 to 5 followed for a mean of 3.2 years. (From O'Hare AM, Choi AI, Bertenthal D, et al. Age affects outcomes in chronic kidney disease. *J Am Soc Nephrol*. 2007;18:2758–2765.)

observation from one longitudinal study that for patients 65 years and older, the risk of ESKD exceeded the risk of death only when the GFR was ≤15 mL/min/1.73 m^2 (Fig. 19.2).[28] By contrast, it has been reported in another study that a more rapid decrease of eGFR was associated with CKD in persons of a younger age, which in turn is associated with an increase in the 1-year mortality rate.[12]

A large individual-level meta-analysis that included data from 2,051,244 participants in 46 cohort studies has provided robust data regarding the role of age on risks related to CKD. In the general population and high cardiovascular risk cohorts, the increase in relative risk of mortality associated with lower GFR was more pronounced in younger persons, but the increase in absolute risk of death was more pronounced in older persons. Similar trends were observed for the mortality risks associated with albuminuria. By contrast, using data from CKD cohorts, the relative increase in risks of mortality and ESKD associated with eGFR and albuminuria were similar among older and younger patients.[29]

Thus older age is a susceptibility factor for CKD, and the associated increase in risk of death and ESKD with the presence of CKD is observed at all ages. These observations suggest that targeted screening for CKD in older subjects would be a cost-effective strategy, but further studies are required to investigate the extent to which the risks associated with CKD in older adults may be attenuated by intervention. For further discussion of CKD in older age groups, see Chapters 21 and 60.

SEX

Data regarding the role of sex on the risk of CKD and progression are somewhat contradictory. Studies have reported a higher incidence of proteinuria and CKD among males in the general population and an increased risk of ESKD or death in males with CKD,[22,25,30] a higher risk of decline in kidney function in males with hypertension,[26] a lower risk of ESKD in females with CKD stage 3,[27] and a shorter time to renal replacement therapy (RRT) in males with CKD stage 4 or 5.[31] In addition, most national registries including the U.S. Renal Data Service have reported a substantially higher incidence of ESKD in males (413 per million population [pmp] in 2003) compared with females (280 pmp).[32] Previous meta-analyses have yielded conflicting results, with one reporting an accelerated rate of decline in GFR in males[33] and another reporting a higher risk of doubling of serum creatinine or ESKD in females after adjustment for baseline variables including blood pressure and urinary protein excretion.[34]

The largest meta-analysis to date included data from studies of 2,051,158 participants investigating the role of sex on CKD-related outcomes. Whereas the risks of all-cause and cardiovascular mortality were significantly higher in males than females and the relative risk of mortality was higher with lower eGFR and higher albuminuria in both sexes, the gradient was steeper in females than males. The relative risk of ESKD was higher with lower eGFR and higher albuminuria in both sexes, and there was no evidence of a difference in the increase in risk between males and females (Fig. 19.3).[35] One limitation of many of the studies quoted is that the menopausal status of the females was not documented. Nevertheless, it is clear from the most robust data published that CKD is associated with at least the same relative increase in risk of death and ESKD in females as in males. The reasons for the higher absolute incidence of RRT in males versus females require further investigation and may be related to treatment preferences for RRT rather than biological differences in disease progression. For further discussion of the role of sex on CKD, see Chapter 22.

RACE AND ETHNICITY

African Americans are overrepresented in the U.S. dialysis population, suggesting that race may be a strong risk factor for the progression of CKD to ESKD. Population-based studies have found a higher incidence of ESKD among African Americans that was attributable only in part to socioeconomic and other known risk factors.[8,25,36,37] Similarly, the risk of early kidney function decline (increase in serum creatinine ≥0.4 mg/dL) was approximately threefold higher (odds ratio [OR], 3.15; 95% CI, 1.86 to 5.33) among Black versus White adults with diabetes, but 82% of this excess risk was attributable to socioeconomic and other known risk factors.[38] The risk of kidney function decline over 5 years among patients with hypertension was more pronounced in African Americans,[26] and African ancestry was independently associated with an accelerated rate of GFR decline in the Modification of Diet in Renal Disease (MDRD) study.[11] Interestingly, data from the Reasons for Geographic and Racial Differences in Stroke (REGARDS) Cohort Study have shown a lower prevalence of eGFR (50 to 59 mL/min/1.73 m^2) among African American persons but a higher prevalence of eGFR (10 to 19 mL/min/1.73 m^2) among white persons,[39] suggesting that African American race acts as a "progression factor" but not as a "susceptibility factor." A 2012 report from the U.S. Renal Data Service showed a substantially higher incidence of ESKD in African American persons (3.3 times higher than whites), Hispanic persons (1.5 times higher than non-Hispanics), and Native American persons (1.5 times higher than whites).[40] Similarly, the prevalence of ESKD in 2012 was higher among minority groups: African American, 5671 pmp; Native American, 2600 pmp; Hispanic, 2932 pmp; Asian, 2272 pmp; and white, 1432 pmp.[40] CKD and ESKD have also been reported to be more prevalent in other ethnic groups, including Asian,[41] Hispanic,[42] Native American,[43] Mexican American,[44] and Aboriginal Australian.[45] A large meta-analysis that included data from 940,366 participants in 25 general population cohort studies investigated the associations of race and ethnicity with CKD. The absolute risk of all-cause or cardiovascular mortality (after adjustment for age) and ESKD was higher in Black versus White versus Asian persons. However, the relative risk of all-cause or cardiovascular mortality and ESKD increased to a similar degree with lower GFR or higher albuminuria in all the race/ethnicity groups.[46]

Thus the associations among lower GFR or higher albuminuria and mortality or ESKD were not modified by race or ethnicity. The mechanisms underlying the associations between race and ethnicity with CKD remain to be elucidated, but possible explanations include genetic factors (*APOL1* gene variants account for 40%, see the "Hereditary Factors" section); prevalence of DM; nephron endowment; susceptibility to salt-sensitive hypertension; and environmental, lifestyle, and socioeconomic differences. Race and ethnicity with CKD are discussed further in Chapters 18, 20, and 50.

HEREDITARY FACTORS

Hereditary kidney diseases resulting from a single gene defect, such as autosomal dominant polycystic kidney disease (ADPKD), Alport syndrome, Fabry disease, and congenital nephrotic syndrome, account for a relatively small proportion of all patients with CKD. Nevertheless, evidence is rapidly accumulating that genetic factors account for familial clustering of other forms of CKD with multifactorial causes. Among 25,883 patients newly diagnosed with ESKD, 22.8% reported a family history of ESKD,[47] and screening of the relatives of patients with ESKD revealed evidence of CKD in 49.3%.[48] In another case-control study including 689 patients with ESKD and 361 controls, having one first-degree relative with CKD was associated with a (nonsignificant) 30% increase in the risk of ESKD (hazard ratio [HR], 1.3; 95% CI, 0.7 to 2.6) and having two such relatives was significantly associated with a 10-fold increase in risk (HR, 10.4; 95% CI, 2.7–40.2) after controlling for multiple known risk factors including diabetes and hypertension.[49] Similarly, a case-control study of 103 white patients with ESKD in the United States reported a 3.5-fold increase in risk of ESKD (95% CI, 1.5–8.4) with the presence of a first-, second-, or third-degree relative with ESKD.[50] A genetic explanation for the high incidence of ESKD observed in African Americans was provided by groundbreaking research that identified strong associations between ESKD and two coding variants in the apolipoprotein L1 *(APOL1)* gene. These gene variants confer resistance to infection with *Trypanosoma brucei rhodesiense*, the cause of African sleeping sickness. This observation explains how selection likely resulted in a high prevalence of these variants in the population. Subsequent studies

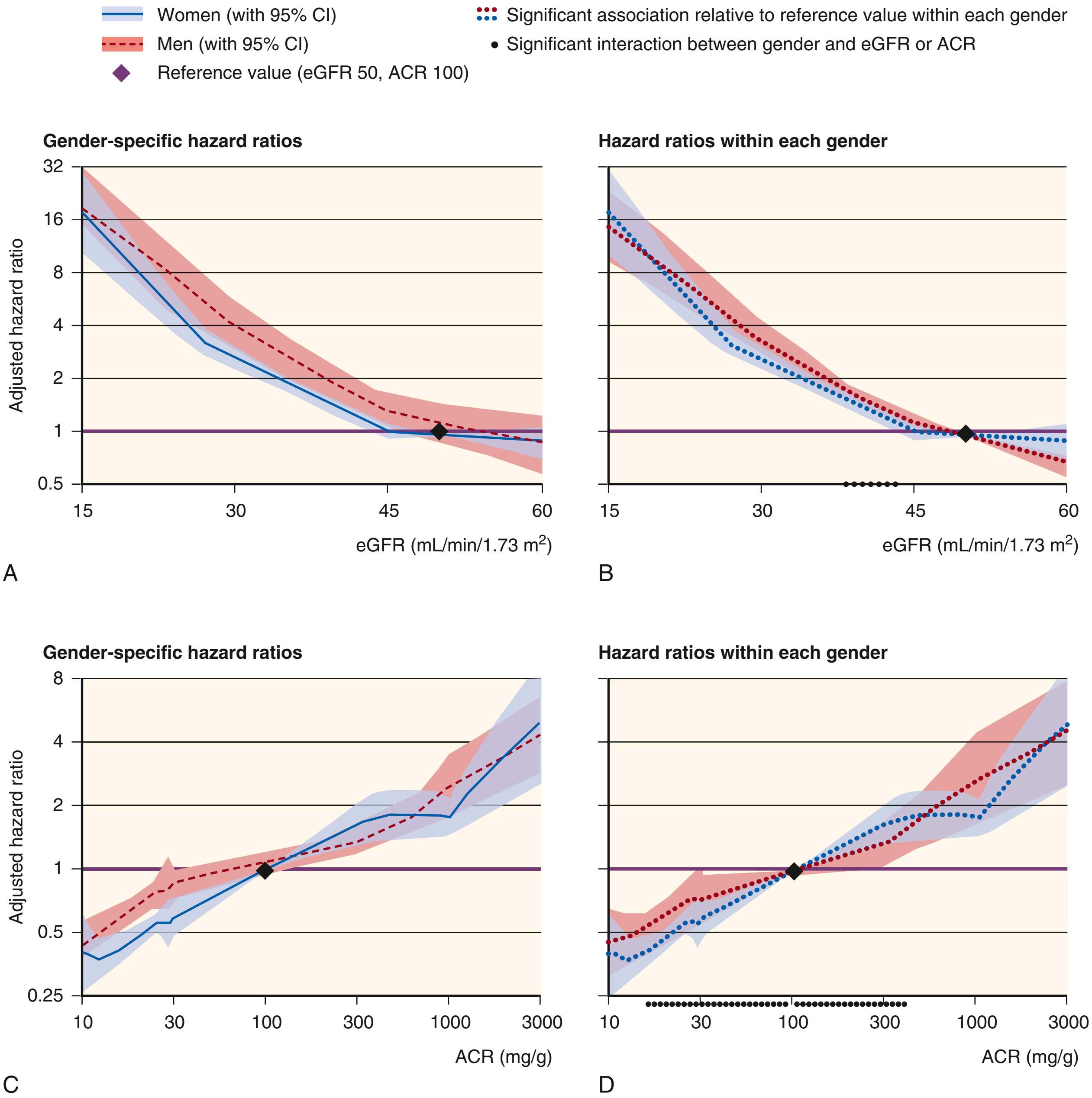

Fig. 19.3 (A and B) Hazard ratios of end-stage kidney disease according to estimated glomerular filtration rate (eGFR). (C and D) Urinary albumin-to-creatinine ratio (ACR) in men versus women in chronic kidney disease cohorts. (A and C) Gender-specific hazard ratios including a main effect for male gender at the reference point. (B and D) Hazard ratios in each gender, thus visually removing the baseline difference between men and women. Hazard ratios were adjusted for age, gender, race, smoking status, systolic blood pressure, history of cardiovascular disease, diabetes, serum total cholesterol concentration, body mass index, and estimated glomerular filtration rate splines or albuminuria. *CI,* Confidence interval. (Adapted from Nitsch D, Grams M, Sang Y, et al. Associations of estimated glomerular filtration rate and albuminuria with mortality and renal failure by sex: a meta-analysis. *BMJ.* 2013;346:f324.)

have identified associations between *APOL1* risk variants and several renal pathologies including focal segmental glomerulosclerosis (FSGS), human immunodeficiency virus–associated nephropathy, sickle cell kidney disease, and severe lupus nephritis.[51] Moreover, cohort studies have reported associations between *APOL1* risk variants and risk of progression to ESKD. Risk of progression was the lowest in European Americans (with no risk variants), intermediate in African Americans with no or one risk variant, and highest in African Americans with two risk variants.[52] It is estimated that *APOL1* variants account for 40% of disease burden due to CKD in African Americans.

Despite the strong association between inheritance of two *APOL1* risk variants and ESKD, only a minority of persons with this genotype actually develop kidney disease, suggesting that the action of a second factor is required to cause disease in genetically susceptible individuals. Human immunodeficiency virus is one example of such a second hit, but it has been proposed that other viruses and other gene variants may also contribute.[51]

Other studies have suggested that genetic factors also increase susceptibility to early manifestations of CKD. In a study of 169 families with one type 2 diabetic proband, the siblings of probands with diabetes and with microalbuminuria were significantly more likely to have microalbuminuria, after adjustment for multiple potential confounders (OR, 3.94; 95% CI, 1.93–9.01), than the siblings of probands with diabetes but without microalbuminuria.[53] Furthermore, the siblings of diabetic probands without diabetes but with microalbuminuria had a significantly higher urinary albumin excretion rate (within the normal range) than the siblings of probands with diabetes and without microalbuminuria.

Genome-wide association studies (GWASs) have identified multiple novel loci that are significantly associated with serum creatinine concentrations or CKD.[54-56] Furthermore, a recent GWAS meta-analysis conducted in 63,558 participants of European descent identified significant associations among GFR decline over time and three gene loci: uromodulin *(UMOD)* (previously associated with CKD and ESKD), *GALNTL5/GALNT11,* and *CDH23.* It was estimated that the heritability of GFR decline in this population was 38%.[57] Further studies have investigated the role of epigenetic factors (heritable changes in the pattern of gene expression not attributable to changes in the primary nucleotide sequence) that may affect the risk of CKD progression. One study compared the genome-wide DNA methylation profile in 20 participants enrolled in the Chronic Renal Insufficiency Cohort (CRIC) study who had exhibited the most rapid decline in GFR and 20 participants with the most stable GFR. Results identified differences in the methylation of several genes associated with epithelial-to-mesenchymal transition and inflammation that may be involved in CKD progression.[58]

From this discussion, it is clear that genetic factors may act as susceptibility factors in some patients, initiating factors in patients with CKD due to a single gene defect, or progression factors in others. The rapid growth in knowledge of genetic aspects of CKD will likely result in genetic risk factors refining risk prediction for patients with CKD. For a more detailed discussion of genetic aspects of kidney disease, see Chapter 44.

HEMODYNAMIC FACTORS

Experimental studies have shown that glomerular hemodynamic responses (e.g., glomerular capillary hypertension, and hyperfiltration) to nephron loss[59] and chronic hyperglycemia[60] are critical factors in establishing the vicious cycle of nephron loss characteristic of CKD. In addition, any factor that increases glomerular hypertension and/or hyperfiltration may be expected to exacerbate glomerular damage and accelerate the progression of CKD (see Fig. 19.1 and Chapter 50).

DECREASED NEPHRON NUMBER

NEPHRON ENDOWMENT

Autopsy studies have revealed that the number of nephrons per kidney varies widely in humans, from 210,332 to 2,702,079 in one series.[61] Multiple factors have been shown to influence nephron endowment including those that affect the intrauterine environment, as well as genetic factors.[62] A substantial body of evidence supports the hypothesis that low nephron endowment predisposes individuals to CKD by provoking an increase in SNGFR and therefore a reduction in renal reserve. The ascertainment of nephron number in living human subjects is currently not possible, but autopsy studies have shown an association between reduced nephron number and hypertension,[63] as well as glomerulosclerosis.[64] In human autopsy studies, birth weight was directly correlated with nephron number[65,66]; therefore birth weight may serve as a proxy of nephron endowment. Low birth weight is also a risk factor for later-life hypertension and DM, both of which increase the risk of CKD.[67] One meta-analysis of 32 studies, which included data from >2 million persons, reported a significantly increased risk of albuminuria (OR, 1.81; 95% CI, 1.19–2.77) and ESKD (OR, 1.58; 95% CI, 1.33–1.88) associated with low birth weight.[68] Low birth weight may be regarded as a susceptibility and progression risk factor for CKD. Factors affecting nephron endowment and the consequences of reduced nephron endowment are discussed in more detail in Chapter 20.

ACQUIRED NEPHRON DEFICIT

In experimental models of acquired nephron deficit, severe nephron loss (5/6 nephrectomy) alone initiates a cycle of progressive injury in the remaining glomeruli, mediated primarily through glomerular hypertension and hyperfiltration.[59] In 14 patients subjected to similarly large reductions in nephron number following partial resection of a single kidney, 2 developed ESKD and 9 developed proteinuria, the extent of which was inversely correlated with the amount of renal tissue remaining.[69] Lesser degrees of acquired nephron loss, such as removal of one of two previously normal kidneys (uninephrectomy), may not be sufficient to cause CKD in most patients.[17,70,71] However, nephrectomy for renal cell carcinoma results in an increased risk of developing CKD that is higher after radical nephrectomy than partial nephrectomy, suggesting that in the presence of subclinical kidney damage, acquired nephron loss may provoke CKD, and that the risk is proportional to the number of functional nephrons removed.[72]

Nephron loss may also predispose individuals to CKD if they are also exposed to other risk factors. This is perhaps best illustrated by the observation that uninephrectomy exacerbates renal injury in experimental diabetic nephropathy[73] and, in persons with diabetes, uninephrectomy increases the risk of developing diabetic nephropathy.[19]

The interaction between nephron loss and other risk factors is further illustrated by the observation that in a study of 488 patients who had surgery for renal cell carcinoma, radical nephrectomy (compared with partial nephrectomy), diabetes, and older age were each independently associated with an increased risk of developing CKD at least 6 months after surgery. In patients who underwent partial nephrectomy but who had no additional risk factors, only 7% developed CKD, but the proportion with CKD was 24%, 30%, and 42% in those 60 years or older, those with hypertension, and those with diabetes, respectively.[74]

In most forms of CKD, initial nephron loss due to primary kidney disease, multisystem disorders that involve the kidney, or exposure to nephrotoxins is focal, but hemodynamic adaptations in the remaining glomeruli are thought to contribute to nephron loss by provoking further glomerulosclerosis (see Chapter 50). Multiple epidemiologic studies support this hypothesis by showing that patients with a reduced GFR are at increased risk of a further decline in kidney function. Several large meta-analyses of cohort studies have identified baseline eGFR as an independent predictor of ESKD. Among 845,125 participants from the general population, eGFR was independently associated with the risk of ESKD when it fell below 75 mL/min/1.73 m^2. Similar findings were reported in 173,892 participants selected on the basis of CKD risk (Fig. 19.4).[75] In the largest meta-analysis to date that included more than 27 million participants, eGFR was an independent risk factor for kidney failure requiring replacement therapy (and multiple other adverse outcomes).[76] Other analyses by the CKD Prognosis Consortium have confirmed that the association between eGFR and risk of ESKD persists independently of sex,[35] age,[29] race and ethnicity,[46] diabetes,[77] and hypertension.[78] In addition, analysis of data from 1,530,648 participants has shown that change in eGFR over time is strongly predictive of future risk of ESKD (and mortality), suggesting that an eGFR decline of 30% may be useful as a surrogate marker of CKD progression in clinical trials.[79] Thus in different contexts, an acquired nephron deficit may be regarded as a susceptibility factor (e.g., after donor nephrectomy in a healthy kidney donor), initiation factor (when severe nephron loss provokes glomerulosclerosis in remaining previously normal glomeruli), or progression factor (when nephron loss accelerates preexisting damage in remaining glomeruli).

ACUTE KIDNEY INJURY

Despite previous perceptions that patients who recover from acute kidney injury (AKI) regain normal kidney function and have a good prognosis, several cohort studies have reported that recovery from AKI is associated with a substantially increased risk of CKD and death. Among 3769 adults who required dialysis for AKI and survived dialysis free for at least 30 days, the incidence rate for maintenance dialysis was 2.63/100 versus 0.91/100 person years in 13,598 matched controls (adjusted HR, 3.23; 95% CI, 2.70–3.86).[80] The relative risk was particularly high for those with no previous diagnosis of CKD (adjusted HR, 15.54; 95% CI, 9.65–25.03). There was no difference in survival between the groups. In another study of similar design, outcomes were investigated in 343 patients with a preadmission eGFR >45 mL/min/1.73

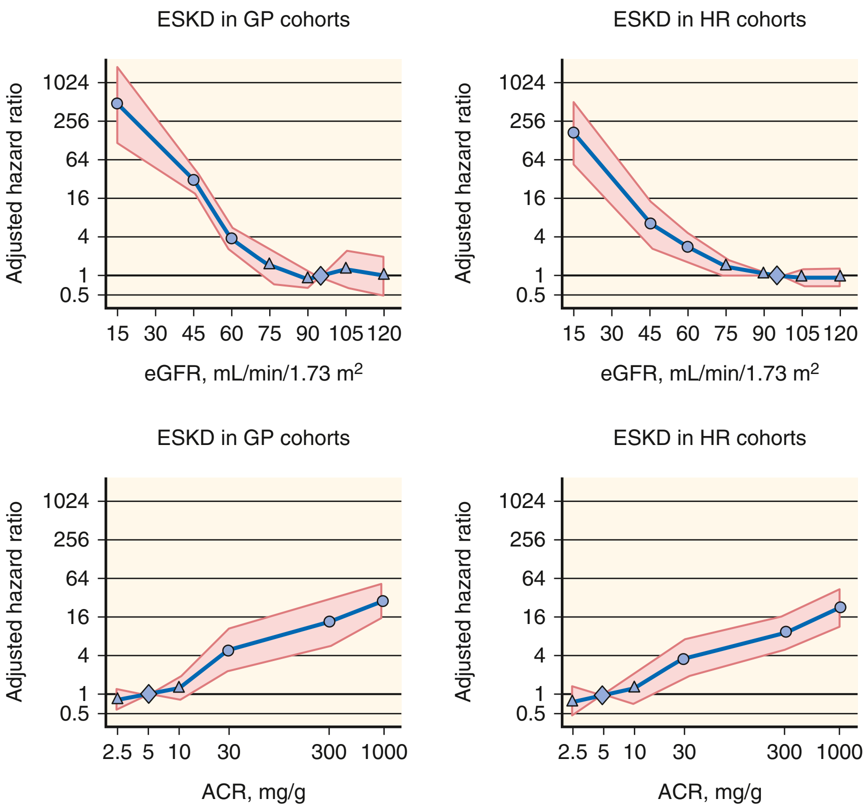

Fig. 19.4 Pooled hazard ratios (95% confidence interval) for end-stage kidney disease (ESKD) according to spline estimated glomerular filtration rate (eGFR) *(upper panels)* and albumin-to-creatinine ratio *(lower panels),* adjusted for each other and for age, gender, and cardiovascular risk factors (continuous analyses). Reference categories are eGFR of 95 mL/min/1.73 m^2 and albumin-to-creatinine ratio of 5 mg/g or dipstick negative or trace. *(Left panels)* Results for general population cohorts. *(Right panels)* High-risk cohorts. Dots represent statistical significance, triangles represent nonsignificance, and shaded areas are 95% confidence interval. *ACR,* Albumin-to-creatinine ratio; *GP,* general population; *HR,* high risk. (From Gansevoort RT, Matsushita K, van der Velde M, et al. Chronic Kidney Disease Prognosis Consortium: lower estimated GFR and higher albuminuria are associated with adverse kidney outcomes in both general and high-risk populations. A collaborative meta-analysis of general and high-risk population cohorts. *Kidney Int*. 2011;79:1341–1352.)

m^2 who required dialysis for AKI but survived for at least 30 days after discharge without dialysis. After controlling for potential confounders, AKI that required dialysis was associated with a 28-fold increase in the risk of developing CKD stage 4 or 5 (adjusted HR, 28.1; 95% CI, 21.1–37.6) and more than double the risk of death (adjusted HR, 2.3; 95% CI, 1.8–3.0) versus 555,660 adult patients hospitalized during the same period but without AKI.[81]

Analysis of data from a cohort of 233,803 Medicare beneficiaries 67 years or older who were hospitalized in 2000 reported a substantially increased risk of developing ESKD in those who developed AKI on a background of CKD (HR, 41.2; 95% CI, 34.6–49.1) or without previous CKD (HR, 13.0; 95% CI, 10.6–16.0) versus those who did not develop AKI. The importance of AKI as a risk factor for CKD initiation was further illustrated by the observation that among patients who had AKI without preexisting CKD (N = 4730), 72.1% developed CKD within 2 years of the AKI episode. Furthermore, roughly one in four (25.2%) of patients with incident-treated ESKD had a history of AKI.[82] In a cohort study of 113,272 patients hospitalized with a primary diagnosis of acute tubular necrosis (ATN), AKI, pneumonia, or acute myocardial infarction (MI) (control group), 11.4% progressed to CKD stage 4 during follow-up including 20.0% of those with ATN, 13.2% of those with AKI, 24.7% of those with preexisting CKD, and 3.3% of patients with acute MI. After controlling for other variables, having a diagnosis of AKI, ATN, or CKD increased the risk of developing CKD stage 4 by 303%, 564%, and 550%, respectively, versus controls. After controlling for covariates, AKI and CKD were associated with an increased risk of death of 12% and 20%, respectively, versus controls.[83]

The multiplicative effect of AKI on CKD progression is further illustrated by a study of 39,805 patients with an eGFR <45 mL/min/1.73 m^2 before hospitalization. Those who survived an episode of dialysis-requiring AKI had a high risk of developing ESKD within 30 days of hospital discharge (i.e., nonrecovery of AKI) that was related to the preadmission eGFR. For an eGFR of 30 to 44 mL/min/1.73 m^2, the incidence of ESKD was 42%, and for an eGFR of 15 to 29 mL/min/1.73 m^2, it was as high as 63%, whereas the incidence of ESKD was only 1.5% in those who did not have dialysis-requiring AKI. In patients who survived longer than 30 days after hospital discharge without ESKD, the incidence of ESKD and death at 6 months was 12.7% and 19.7%, respectively, versus 1.7% and 7.4% in the comparator group with CKD but no AKI. After adjustment for multiple risk factors, AKI was associated with a 30% increase in long-term risk for death or ESKD (adjusted HR, 1.30; 95% CI, 1.04–1.64).[84]

Consistent with the findings of individual studies, a meta-analysis of 13 cohort studies reported a significantly increased risk of developing CKD and ESKD in patients who had survived an episode of AKI versus participants without AKI (pooled adjusted HR for CKD, 8.8; 95% CI, 3.1–25.5; pooled HR for ESKD, 3.1; 95% CI, 1.9–5.0).[85] Taken together, these data show that AKI should be regarded as an important risk factor for CKD initiation and progression. The mechanisms responsible for these observations require further elucidation but have been proposed to include nephron loss, loss of peritubular capillaries, cell cycle arrest, cell senescence, pericyte and myofibroblast activation, fibrogenic cytokine production, and interstitial fibrosis.[86,87]

HYPERTENSION

Hypertension, which is defined as an increased systolic blood pressure >130 mm Hg and/or diastolic blood pressure >80 mm Hg, or the need for pharmacologic therapy to achieve BP targets, is an almost universal consequence of impaired function.[88] From a CKD-focused standpoint, this is due to sodium retention, hypervolemia, and neurohormonal activation associated with progressive decrease in GFR.[89-91] However, hypertension itself is also known to be an important factor in the progression of CKD toward ESKD. In the hypothesis of CKD progression presented in Fig. 19.1, it is clear that elevated systemic blood pressure transmitted to the glomerulus would contribute to glomerular hypertension and thus accelerate glomerular damage. Hypertension has been shown to be associated with ESKD in several large population-based studies.[22,23,30,92] Furthermore, a close association between the magnitude of increased risk and level of blood pressure has been reported in several studies, so even elevations in blood pressure below the threshold for the diagnosis of hypertension were associated with increased risk of ESKD.[22,30,93]

Among patients with CKD in the MDRD study, higher baseline mean arterial pressure (MAP) was independently associated with a more rapid rate of GFR decline.[11] These observations have led to the suggestion that blood pressure be viewed as a continuous rather than dichotomous risk factor for CKD, with less emphasis on traditional definitions of hypertension and normotension.[94] Whereas the primary analysis of data from the MDRD study found no significant difference between the rate of decline in GFR among patients randomized to intensive blood pressure control (target MAP <92 mm Hg, equivalent to <125/75 mm Hg) and standard blood pressure control (target MAP < 107 mm Hg, equivalent to 140/90 mm Hg), a secondary analysis did show benefit associated with the lower blood pressure target in patients with higher levels of baseline proteinuria.[95] Further secondary analysis showed that the lower *achieved* blood pressure was also associated with a slower GFR decline, an effect that was more marked in patients with higher baseline proteinuria.[96] Thus the MDRD study results suggest a significant interaction between blood pressure and proteinuria as risk factors for CKD progression.

The Systolic Blood Pressure Intervention Trial, which randomized nearly 10,000 patients older than 50 years of age with hypertension, without diabetes, and increased risk of CKD including about one-third with CKD, showed that targeting systolic blood pressure <120 mm Hg, as compared with <140 mm Hg, resulted in fewer fatal and nonfatal CVEs and death.[97] A subgroup analysis of the 2646 participants with CKD showed no difference in the primary renal outcome of ≥50% decline in GFR or ESKD, though there was a more pronounced decrease in GFR after 6 months (–0.47 vs. –0.32 mL/min/1.73 m^2 per year) among patients randomized to the lower SBP target. The risks of all-cause mortality (HR, 0.72; 95% CI, 0.53–0.99) and CVE (HR, 0.81; 95% CI, 0.63–1.05) were lower in the more intensive SBP-lowering group and there was no overall increase in serious adverse events, though AKI, hypokalemia, and hyperkalemia were increased in the more intensive SBP-lowering target group.[98] An additional analysis of two intensive systolic blood pressure lowering trials also found a threefold increase in incident CKD among patients without CKD at baseline, although no change in tubular injury markers was observed.[99] In the HALT Progression of Polycystic Kidney Disease (HALT-PKD) study, targeting a blood pressure of ≤120/70 mm Hg in patients with autosomal dominant polycystic kidney disease (ADPKD) through use of lisinopril and telmisartan allowed for a slower increase in total kidney volume, no overall change in eGFR, a more pronounced decline in left ventricular mass index, and a more pronounced reduction in albuminuria.[100]

Taken together, there is convincing evidence that elevated blood pressure is a risk factor for progression of CKD. While lowering SBP reduces the risk of CKD and death, it is unknown whether more intensive control of hypertension reduces (or increases) the risk of progressive CKD.

OBESITY AND METABOLIC SYNDROME

In experimental models, obesity is associated with hypertension, proteinuria, and progressive kidney disease. Micropuncture studies have confirmed that obesity is another cause of glomerular hyperfiltration and glomerular hypertension that can be predicted to exacerbate the progression of CKD.[101,102] Furthermore, several other factors associated with obesity and the metabolic syndrome may contribute to kidney damage including hormones and proinflammatory molecules produced by adipocytes,[103] elevated mineralocorticoid levels and/or mineralocorticoid receptor activation by cortisol,[104] and reduced adiponectin levels.[105] In humans, severe obesity is associated with increased renal plasma flow, glomerular hyperfiltration, and albuminuria, abnormalities that are reversed by weight loss.[106] Obesity, as defined by elevated body mass index (BMI), has been associated with an increased risk of developing CKD in several large population-based studies.[23,107,108] Indeed, one study found a progressive increase in the relative risk of developing ESKD associated with higher BMI (relative risk, 3.57; 95% CI, 3.05 to 4.18 for BMI of 30.0 to 34.9 kg/m^2 vs. BMI of 18.5 to 24.9 kg/m^3) among 320,252 subjects confirmed to have no evidence of CKD at initial screening.[109]

Evidence indicates that obesity may directly cause a specific form of glomerulopathy characterized by proteinuria and histologic features of focal and segmental glomerulosclerosis,[110,111] but it is likely that it also acts as a risk factor in the development of several other forms of kidney disease. One study has identified childhood obesity as a risk factor for CKD in adulthood. Among 4340 participants born in 1 week in 1946, pubertal-onset obesity and obesity throughout childhood were associated with an increased risk of CKD (defined by eGFR <60 mL/min/1.73 m^2 or albuminuria) at age 60 to 64 years.[112] Interest has also focused on the role of metabolic syndrome (insulin resistance), as defined by the presence of abdominal obesity, dyslipidemia, hypertension, and fasting hyperglycemia, in the development of CKD. An analysis of data from the Third National Health and Nutrition Examination Survey (NHANES III) data found a significantly increased risk of CKD and microalbuminuria in survey participants with the metabolic syndrome, as well as a progressive increase in risk associated with the number of components of the metabolic syndrome present.[113] Furthermore, a longitudinal study of 10,096 patients without diabetes or CKD at baseline identified metabolic syndrome as an independent risk factor for the development of CKD over 9 years (adjusted OR, 1.43; 95% CI, 1.18–1.73). Again, there was a progressive increase in risk associated with the number of traits of the metabolic syndrome present (OR, 1.13; 95% CI, 0.89–1.45 for one trait vs. OR, 2.45; 95% CI, 1.32–4.54 for five traits).[114] In another study, patient hip-to-waist ratio, a marker of insulin resistance, was independently associated with impaired kidney function, even in lean individuals (BMI <25 kg/m^3), among a population-based cohort of 7676 individuals.[115]

Surgical intervention in the form of gastric banding or bypass appears to have evidence in improving long-term outcomes. Beneficial renoprotective effects of weight loss have been reported in a meta-analysis of observational studies that found an association between weight loss and reduction in proteinuria independent of blood pressure,[116] as well as smaller studies that reported improvement or stabilization of renal function[117] or reduction in proteinuria[118] following bariatric surgery in subjects with CKD.

In recent years, there has been an accumulation of evidence that treatment of obesity in patients with or without diabetes with GLP1 receptor agonists can improve kidney and cardiovascular outcomes. Previous CV outcome trials of liraglutide, dulaglutide, and semaglutide that included secondary analyses of kidney outcomes showed GLP1 receptor agonists reduced major adverse cardiovascular events (MACE) by 14% (HR, 0.86; 95% CI, 0.80–0.93) and hospital admissions for heart failure by 11% (HR, 0.89; 95% CI, 0.82–0.98). Additionally, the risk of various kidney disease outcomes including macroalbuminuria, increased serum creatinine, a ≥40% decline in eGFR, receival of kidney replacement therapy, or death due to kidney disease was reduced by 21% (HR, 0.79; 95% CI, 0.73–0.87).[119] Findings from the FLOW trial were published in May 2024 and conclusively demonstrated a benefit of semaglutide, such that major kidney disease events including onset dialysis, transplantation, reduction of eGFR to <15 mL/min/1.73 m^2, or death from kidney- or cardiovascular-related causes was reduced by 24% (HR, 0.76; 95% CI, 0.66–0.88) after weekly 1.0 mg subcutaneous dose of semaglutide. Further, the risk of MACE was reduced by 18% (HR, 0.82; 95% CI, 0.68–0.98).[120] This evidence supports the measurement/evaluation of obesity in patients with CKD and suggests that medical therapy with GLP1 agonists is likely to modify both kidney and cardiovascular risk.

The best method for assessing obesity in CKD remains to be determined. A systematic review analyzed the effects of weight loss achieved by bariatric surgery, medication, or diet in 31 studies and found that in most studies, weight loss was associated with reductions in proteinuria. In people with glomerular hyperfiltration, the GFR tended to decrease with weight loss, and in those with a reduced GFR, it tended to increase.[121] BMI is the most widely applied method but does not take body composition into account. One study has reported a high sensitivity but relatively low specificity of BMI to detect obesity in subjects with CKD.[122]

HIGH DIETARY PROTEIN INTAKE

Consistent with the hypothesis that the glomerular hemodynamic changes associated with hyperfiltration accelerate glomerular injury, experimental studies have reported that a high-protein diet accelerates kidney disease progression, whereas dietary protein restriction[123,124] results in normalization of glomerular capillary hydraulic pressure, as well as SNGFR and marked attenuation of glomerular damage.[59] Observational studies in humans have reported an increased risk of microalbuminuria associated with higher dietary protein intake in patients with diabetes and hypertension (OR, 3.3; 95% CI, 1.4–7.8) but not in healthy subjects or those with isolated diabetes or hypertension,[125] again illustrating the interaction between risk factors for CKD. In another study, high dietary protein intake, particularly nondairy animal protein, was associated with a more rapid rate of GFR decline among women with an eGFR of 80 to 55 mL/min/1.73 m^2 but not in those with an eGFR of >80 mL/min/1.73 m^2.[126] Randomized trials investigating the effects of high-protein diet are lacking, but several studies have sought to examine the potential renoprotective effects of dietary protein restriction. In the MDRD study, the primary analysis revealed no significant difference in the mean rate of GFR decline in subjects randomized to low- or very-low protein diets,[95] but secondary analysis of outcomes according to *achieved* dietary protein intake indicated that a reduction in protein intake of 0.2 g/kg/day correlated with a 1.15-mL/min/year reduction in the rate of GFR decline, equivalent to a 29% reduction in the mean rate of GFR decline.[127] By contrast, long-term follow-up of participants in study 2 of the MDRD trial found no renoprotective benefit among those randomized to very-low protein diet in the original study but reported a higher risk of death in this group (HR, 1.92; 95% CI, 1.15–3.20).[128] As such, dietary protein restriction cannot be routinely recommended for patients with CKD. Additional discussion on the role of dietary protein restriction in the management of CKD is available in Chapter 55.

PREGNANCY AND PREECLAMPSIA

Physiologic adaptations during pregnancy provoke glomerular hyperfiltration that usually does not result in kidney damage. In the context of preexisting CKD, however, the glomerular hyperfiltration of pregnancy can be predicted to exacerbate proteinuria and glomerular injury. Several studies have shown an increased risk of CKD progression during pregnancy, particularly when the pregestational serum creatinine is ≥1.4 mg/dL (≥124 μmol/L). In one study of 82 pregnancies in 67 females with primary kidney disease and serum creatinine concentrations ≥1.4 mg/dL, blood pressure, serum creatinine, and proteinuria increased during pregnancy. In 70 pregnancies with postpartum data available, persistent loss of maternal kidney function at 6 months was reported in 31%, and by 12 months eight (14%) females had progressed to ESKD. Adverse obstetric outcomes included preterm delivery in 59% and low birth weight in 37%, although fetal survival was 93%.[129]

In a more recent series of 49 females with CKD stage 3 to 5 before pregnancy, the mean GFR declined during pregnancy (from 35 ± 12.2 to 30 ± 13.8 mL/min/1.73 m^2), but there was no change in the mean postpartum rate of GFR decline. Nevertheless, a pregestational GFR <40 mL/min/1.73 m^2, combined with proteinuria of >1 g/day, was associated with a more rapid postpartum GFR decline and a shorter time to ESKD or halving of GFR and low birth weight.[130] Although earlier reports suggested good outcomes, one study has reported adverse pregnancy outcomes associated even with early stage CKD. In 91 pregnancies with predominantly CKD stages 1 and 2, modest increases in hypertension, serum creatinine, and proteinuria were observed.[131] An increase in adverse pregnancy outcomes including preterm delivery, lower birth weight, and admission to a neonatal intensive care unit versus low-risk pregnancy controls was also reported; this remained true, even when only those with CKD stage 1 (GFR >90 mL/min/1.73 m^2) were considered, although there were no perinatal deaths. By contrast, pregnancy was not associated with a more rapid decline in the GFR over 5 years in a cohort of 245 females of childbearing age with IgA nephropathy and serum creatinine level of ≤1.2 mg/dL (in the majority).[132]

Complications of pregnancy and, in particular, hypertension and preeclampsia may also contribute to kidney damage. In one large population-based study, renal outcomes were assessed in 570,433 women who had had at least one singleton pregnancy. Only 477 women developed ESKD at a mean of 17 ± 9 years after the first pregnancy (overall rate, 3.7/100,000 women/year), but preeclampsia was associated with a significant increase in the risk of ESKD, ranging from a relative risk of 4.7 for preeclampsia in a single pregnancy (95% CI, 3.6–6.1) to a relative risk of 15.5 for preeclampsia in two or three pregnancies (95% CI, 7.8–30.8). The risk was further increased if the pregnancy resulted in a low-birth-weight or preterm infant. Causes of ESKD were glomerulonephritis in 35%, hereditary or congenital disease in 21%, diabetic nephropathy in 14%, and interstitial nephritis in 12%.[133] Similarly, in women with diabetes before pregnancy, preeclampsia and preterm birth were associated with significantly increased risks of ESKD and death, illustrating how different risk factors for CKD may interact to increase risk.[134]

A large cohort study reported an increased risk of multiple adverse health outcomes after hypertension during pregnancy including CVD, DM, and CKD (HR, 1.91; 95% CI, 1.18–3.09).[135] Similarly, a large case-control study found that hypertension during pregnancy was associated with a substantially increased risk of subsequent CKD (HR, 9.38; 95% CI, 7.09–12.4) or ESKD (HR, 12.4; 95% CI, 8.53–18.0).[136] In both these studies, the risks of CKD were substantially higher if preeclampsia developed during the pregnancy. Possible explanations for these observations include the presence of pathogenic factors common to CKD and preeclampsia including obesity, hypertension, insulin resistance, and endothelial dysfunction; exacerbation by preeclampsia of preexisting subclinical CKD; and effects of preeclampsia on the kidney that increase the risk of CKD later in life.[137] That preeclampsia may provoke kidney damage has been suggested by several studies showing an increased incidence of microalbuminuria after preeclampsia. A meta-analysis of seven of these studies reported a 31%

prevalence of microalbuminuria at a weighted mean of 7.1 years after preeclampsia versus 7% in a control group with uncomplicated pregnancies.[138] Further research is required to identify which mechanisms are most relevant, but even without further information, preeclampsia should be regarded as a risk factor for the development and progression of CKD.

MULTISYSTEM DISORDERS

DIABETES MELLITUS

Diabetic nephropathy has rapidly become the single most common cause of ESKD worldwide. Diabetes was associated with a substantially increased risk of ESKD or death associated with CKD in one population-based study of 23,534 participants (HR, 7.5; 95% CI, 4.8–11.7),[30] as well as an increased risk of moderate CKD (estimated creatinine clearance <50 mL/min) in another study of 1428 participants with an estimated creatinine clearance of >70 mL/min at baseline.[139] Evidence that glycemic control is a key risk factor for the development of diabetic nephropathy has been shown in randomized trials that found a reduced risk of developing nephropathy in participants with type 1[140] and type 2[141] diabetes randomized to tight glycemic control. The pathogenesis of diabetic nephropathy is complex and involves multiple mechanisms including glomerular hemodynamic factors,[60,142] advanced glycation end-product formation, generation of reactive oxygen species, and upregulation of profibrotic growth factors and cytokines.[143,144] In at least one study, diabetic nephropathy was associated with more rapid progression to ESKD than other causes of CKD.[31,145,146] Thus diabetes may be regarded as a susceptibility, initiation, and progression risk factor for CKD. For further discussion of the pathogenesis of diabetic nephropathy, see Chapter 41.

PRIMARY KIDNEY DISEASE

Whereas substantial variation in the rate of GFR decline has been observed among patients with a common cause of CKD, there is also evidence that some forms of CKD may provoke more rapid progression than others. In the MDRD study[11] and the Chronic Renal Insufficiency Standards Implementation Study (CRISIS),[145] a diagnosis of ADPKD was associated with a more rapid rate of GFR decline. In several cohort studies, diabetic nephropathy was associated with shorter time to ESKD[31] or a more rapid rate of GFR decline than other diagnoses.[145,146]

CARDIORENAL SYNDROME

Multiple studies have reported that CKD is associated with a substantial increase in the risk of CVD,[147] and it is therefore not surprising that CVD is also associated with an increased risk of CKD. Among hospitalized Medicare beneficiaries, the prevalence of CKD stage 3 or worse was 60.4% among patients with heart failure and 51.7% among those with a history of MI. The presence of CKD in addition to heart disease was associated with a significantly increased risk of death and progression to ESKD.[148] These observations may in part be explained by the fact that CVD and CKD share many risk factors including obesity, metabolic syndrome, hypertension, DM, dyslipidemia, and smoking. In addition, CVD may exert effects on the kidneys that promote the initiation and progression of CKD including decreased renal perfusion in heart failure and atherosclerosis of the renal arteries. This is a phenomenon known as cardiorenal syndrome (CRS), which is an acute or chronic dysfunction in one organ system leading to dysfunction in the other, specifically in the cardiovascular and renal systems. Mechanisms that lead to the development of CRS include neurohormonal activation, altered renal blood flow, renal congestion, and right ventricular dysfunction.[149] A typical example of CRS is seen in renal atherosclerosis, which was detected in 39% of patients (≥70% stenosis in 7.3%) undergoing elective coronary angiography.[150] Furthermore, arterial stiffness may result in greater transmission of elevated systemic blood pressure to glomerular capillaries and exacerbate glomerular hypertension. In one study, pulse wave velocity (PWV) and augmentation index, markers of arterial stiffness, were identified as independent risk factors for progression to ESKD among patients with CKD stage 4 or 5[151]; in another study, augmentation index was an independent determinant of rate of creatinine clearance decline among patients with CKD stage 3.[152] By contrast, neither PWV nor augmentation index was associated with the rate of GFR decline in a cohort of patients with CKD stages 2 to 4.[153] In two relatively small cohort studies of those with CKD, a diagnosis of CVD was associated with an increased risk of progression to ESKD[154,155] but, in the CRIC study, a history of any CVD at baseline was not associated with the risk of the primary composite endpoint (ESKD or 50% decline in eGFR) among 3939 participants. Conversely, in the same study, a history of heart failure was independently associated with a 29% higher risk of the primary outcome.[156] For further discussion of CVD in patients with CKD and CRS, see Chapter 42.

CONVENTIONAL BIOMARKERS

Biomarkers are parameters that are measured and evaluated to differentiate between normal and abnormal biological processes or predict adverse outcomes. For CKD, biomarkers may be considered a reflection of renal function impairment. Some of the standard biomarkers indicated in CKD progression include proteinuria and several routine serum biomarkers that are essential for evaluation of persons with reduced GFR.[97] The more novel biomarkers are serum and urinary substances that allow for more targeted risk prediction in CKD.[149]

PROTEINURIA

Proteinuria is an indicator of dysfunction in the glomerular filtration barrier and is therefore a marker of glomerulopathy and an index of disease severity. Proteinuria is often a result of glomerular injury and is associated with hyperfiltration, intraglomerular hypertension, glomerular hypertrophy, and glomerulosclerosis.[149] Experimental evidence has suggested that proteinuria may also contribute to progressive kidney damage in CKD (see Chapter 29). A large body of evidence attests to a strong association between proteinuria and the risk of CKD progression, a relation mediated through several mechanisms including tubular injury, inflammation, and fibrosis, as well as cardiovascular and all-cause mortality. Mass screening of a general population of 107,192 participants by dipstick urinalysis identified proteinuria as the most powerful predictor of ESKD risk over 10 years (OR, 14.9; 95% CI, 10.9–20.2).[22]

Similarly, among 12,866 middle-aged men enrolled in the Multiple Risk Factor Intervention Trial (MRFIT), proteinuria detected by dipstick test was associated with a significantly increased risk of developing ESKD over 25 years (HR for 1+ proteinuria, 3.1; 95% CI, 1.8–3.8; HR for ≥2+ proteinuria, 15.7; 95% CI, 10.3–23.9). Furthermore, detection of 2+ proteinuria or more increased the hazard ratio for ESKD associated with an eGFR <60 mL/min/1.73 m^2 from 2.4 without proteinuria (95% CI, 1.5–3.8) to 41.0 with proteinuria (95% CI, 15.2–71.1).[157] Similar associations have been reported for measurements of urinary albumin in the general population. In the Nord-Trøndelag Health (HUNT 2) study, which included 65,589 adults, microalbuminuria and macroalbuminuria were independent predictors of ESKD after 10.3 years (HR, 13.0 and 47.2, respectively) and combining reduced eGFR with albuminuria substantially improved the prediction of ESKD.[158]

Among patients selected for having CKD from a wide variety of causes, baseline proteinuria has consistently predicted adverse kidney outcomes.[159-161] In three large prospective studies that included patients with nondiabetic CKD (MDRD study, Ramipril Efficacy In Nephropathy [REIN] study, and AASK), higher baseline proteinuria was strongly associated with a more rapid decline in GFR.[11,96,162,163] Similarly, among patients with diabetic nephropathy, the baseline urinary albumin-to-creatinine ratio (UACR) was an independent predictor of ESKD in the Reduction of Endpoints in NIDDM with the Angiotensin II Antagonist Losartan (RENAAL) study and Irbesartan in Diabetic Nephropathy Trial (IDNT).[164,165] The findings of these individual studies have been confirmed by several large meta-analyses. In one analysis, which included nine general population cohorts (N = 845,125) and eight cohorts with increased risk of developing CKD (N = 173,892), UACR of >30, 300, and 1000 mg/g were independently associated with progressive increases in the risk of ESKD, progressive CKD, and AKI, respectively (see Fig. 19.4).[75] Among 21,688 patients known to have CKD from 13 studies, an 8-fold higher UACR or protein-to-creatinine ratio (UPCR) was associated with all-cause mortality (pooled HR, 1.40) and risk of ESKD (pooled HR, 3.04).[166] In a meta-analysis that included more than 9 million participants, UACR was independently associated with kidney failure requiring kidney replacement therapy (and several other adverse outcomes).[76] Other meta-analyses conducted by the CKD Prognosis Consortium have shown that the magnitude of proteinuria remains a risk factor for ESKD independent of sex,[35] race and ethnicity,[46] age,[29] diabetes,[77] or hypertension.[78] Recognition of the importance of proteinuria as a risk factor in CKD prompted the addition of an albuminuria category (A1 to A3) to the CKD classification proposed by KDIGO.[10]

SERUM ALBUMIN

Serum albumin concentrations are widely regarded as a marker of nutritional status but may also be reduced due to inflammation or in the setting of high-grade proteinuria. Several studies have identified lower serum albumin concentrations as a risk factor for CKD progression. In the MDRD study, higher baseline serum albumin was associated with slower subsequent rate of GFR decline, but in a multivariable analysis, serum albumin was displaced by a similar correlation with baseline serum transferrin levels, another marker of protein nutrition.[11] Three studies have found associations between serum albumin and kidney outcomes in patients with type 2 diabetes and CKD. Among 182 patients with a mean serum creatinine of 1.5 mg/dL at baseline, hypoalbuminemia was an independent risk factor for ESKD.[167] In a long-term follow-up of 343 patients, lower baseline serum albumin was an independent predictor of CKD progression and,[168] in the RENAAL study, lower serum albumin was an independent predictor of ESKD.[164] Similar observations have been reported in other forms of CKD. In a large cohort of patients with IgA nephropathy (N = 2269), lower serum total protein (composed largely of albumin) was an independent risk factor for ESKD.[169] In these studies, the predictive value of serum albumin was seen after adjustment for urine albumin (or protein) excretion, suggesting that variation in serum albumin related to inflammation, nutritional status, and/or liver disease also contributed to ESKD risk.

ANEMIA

Chronic anemia due to inherited hemoglobinopathy is associated with increased renal plasma flow, glomerular hyperfiltration, and subsequent development of proteinuria, hypertension, and ESKD.[170,171] Anemia is a common complication of CKD from any cause, and several studies have shown anemia to be an independent predictor of CKD progression. In the RENAAL study, baseline hemoglobin was a significant independent predictor of ESKD among patients with diabetic kidney disease—each 1 g/dL lower hemoglobin concentration was associated with an 11% increase in the risk of ESKD.[172] Baseline hemoglobin was also one of four variables included in the renal risk score developed from the RENAAL data.[164] Similarly, a higher hemoglobin concentration was independently associated with a lower risk of progression to ESKD (halving of GFR or need for dialysis) or death among 131 patients with all forms of CKD (HR, 0.78; 95% CI, 0.64–0.95 for each 1 g/dL increase).[161] Furthermore, time-averaged hemoglobin of <12 g/dL was associated with a significantly increased risk of ESKD among 853 male veterans with CKD stages 3 to 5 (HR, 0.74; 95% CI, 0.65–0.84 for each 1 g/dL higher hemoglobin).[173]

Consistent with the hypothesis that anemia contributes directly to CKD progression, two small randomized studies have reported a renoprotective benefit associated with erythropoietin therapy. Among patients with serum creatinine of 2 to 4 mg/dL and hematocrit <30%, erythropoietin treatment was associated with significantly improved renal survival.[174] In patients without diabetes with serum creatinine of 2 to 6 mg/dL, early treatment (started when hemoglobin <11.6 g/dL) with epoetin alfa was associated with a 60% reduction in the risk of doubling of serum creatinine, ESKD, or death versus delayed treatment (started when hemoglobin <9.0 g/dL).[175] By contrast, two other studies that had left ventricular mass as their primary endpoint[176,177] and the Trial to Reduce Cardiovascular Events with Aranesp Therapy (TREAT)[178] found no effect of a higher versus lower hemoglobin target on the rate of decline in the GFR. Several studies have, however, reported adverse outcomes associated with normalization of hemoglobin in patients with CKD. In the Cardiovascular Risk Reduction by Early Anemia Treatment with Epoetin Beta (CREATE) study, randomization to a higher hemoglobin target (13–15 mg/dL) was associated with a shorter time to initiation of dialysis than a lower target (10.5–11.5 mg/dL).[179] In TREAT, randomization to a higher hemoglobin target was associated with an increased risk of stroke[178] and, in the Correction of Hemoglobin and Outcomes in Renal Insufficiency (CHOIR) study, a higher hemoglobin target was associated with an increased incidence of the combined endpoint of all-cause mortality, MI, or hospitalization for congestive cardiac failure.[180] Large randomized trials with a different class of anemia treatments (HIF stabilizers) have also found no benefit in slowing progression of kidney disease when compared with placebo or active controls.

DYSLIPIDEMIA

Lipid abnormalities are common in patients with CKD, and several studies have identified dyslipidemia as a susceptibility and progression factor for CKD. In population-based studies, several lipid profile abnormalities have been associated with an increased risk of developing CKD including an elevated low-density lipoprotein (LDL)-to–high-density lipoprotein (HDL) cholesterol ratio,[181] higher triglyceride and lower HDL cholesterol concentrations,[182] lower HDL cholesterol concentrations[23] and elevated total cholesterol concentrations, low HDL cholesterol concentrations, and elevated total cholesterol-to-HDL cholesterol ratio.[183] In the MDRD study, lower HDL cholesterol concentrations were independently associated with decline in GFR[11]; in a smaller study of patients with CKD, total cholesterol, LDL cholesterol, and apolipoprotein B concentrations were all associated with a more rapid decline in the GFR.[184] Among 223 patients with IgA nephropathy, hypertriglyceridemia was independently associated with CKD progression.[185] Hypercholesterolemia was reported to predict loss of kidney function in patients with type 1 or 2 diabetes[186,187] and, among patients without diabetes, CKD advanced more rapidly in those with hypercholesterolemia and hypertriglyceridemia.[188]

Randomized controlled trials of lipid lowering have produced mixed results with respect to kidney outcomes. Subgroup analysis of a prospective randomized trial of pravastatin treatment in patients with previous MI found that pravastatin slowed the rate of decline in patients with an eGFR <40 mL/min/1.73 m^2, an effect that was more pronounced in participants with proteinuria at baseline.[189] Similarly, in the Heart Protection Study, patients with previous CVD or diabetes randomized to simvastatin treatment had a smaller increase in serum creatinine than those who received placebo.[190] In a placebo-controlled, open-label study, atorvastatin treatment in patients with CKD, proteinuria, and hypercholesterolemia was associated with preservation of creatinine clearance, whereas creatinine clearance declined in patients receiving placebo.[191] By contrast, lipid lowering with fibrates was unassociated with renoprotection in two studies,[181,192] although one study showed a reduced incidence of microalbuminuria in patients with type 2 diabetes receiving fenofibrate.[193]

Analysis of data from a relatively small subgroup of studies with kidney endpoints recorded in a meta-analysis found that among participants with CKD, statin therapy was associated with a reduction in proteinuria but no improvement in creatinine clearance.[194] Furthermore, analysis of data from 3939 participants in the CRIC study found no association between total or LDL cholesterol and the risk of ESKD or 50% reduction in eGFR. Indeed, among participants with proteinuria of <0.2 g/day, higher LDL and total cholesterol were associated with a lower risk of reaching this composite endpoint.[195] The Study of Heart and Renal Protection (SHARP) is the largest randomized controlled trial to investigate the cardiovascular and kidney protective effects of lipid lowering in CKD. Patients with CKD or undergoing dialysis were randomized to treatment with simvastatin + ezetimibe or placebo. In 6245 participants with CKD not requiring dialysis, treatment resulted in an average reduction in LDL cholesterol of 0.96 mmol/L but did not significantly reduce the risk of ESKD or the composite endpoint of ESKD or doubling of serum creatinine.[196] Together, evidence that dyslipidemia is a risk factor for CKD progression remains mixed, with the most recent studies indicating no association. Mechanisms whereby dyslipidemia may contribute to CKD progression are discussed in Chapter 50.

SERUM URATE

Hyperuricemia is a common consequence of CKD and may also contribute to CKD progression. Most, but not all, population-based studies have identified hyperuricemia as an independent risk factor for the development of incident CKD. Similarly, most cohort studies that included people with CKD have identified a higher serum urate as a risk factor for CKD progression. Possible mechanisms whereby hyperuricemia may contribute to CKD progression are exacerbation of glomerular hypertension,[197,198] endothelial dysfunction,[199,200] and proinflammatory effects.[201] By contrast, it is possible that an elevated uric acid concentration is acting as a marker of reduced kidney function or oxidative stress—uric acid is produced by xanthine oxidase, which also generates reactive oxygen species.

Two large randomized trials have examined the efficacy of urate-lowering treatments on CKD progression. These trials examined the effects of different xanthine oxidase inhibitors (allopurinol and febuxostat) and found no effects on CKD progression despite substantial reductions in serum urate concentrations. Thus it appears that while serum urate uric is a marker for impaired kidney function, it is not a modifiable risk factor for slowing CKD progression.[202,203]

METABOLIC ACIDOSIS

Metabolic acidosis commonly develops in patients with CKD, particularly among diseases with extensive tubulointerstitial involvement and more generally with a reduction in functional nephron mass. As a result, there is a subsequent elevation of cortical ammonia levels, renin–angiotensin–aldosterone system activation, complement cascade activation, endothelial/aldosterone activation, and inflammation, all of which lead to impaired renal acid clearance.[204] At least five studies have investigated metabolic acidosis as a risk factor in human CKD. In all except the MDRD study,[205] lower serum bicarbonate concentrations, even within the population reference range, were independently associated with CKD progression.[206-209] Two small randomized trials have reported slowing of CKD progression with bicarbonate supplementation,[210,211] and another trial found that correction of acidosis with oral sodium bicarbonate or a diet rich in fruits and vegetables was associated with a lower rate of GFR decline.[212] Several randomized controlled trials have also indicated that correction of acidosis with bicarbonate supplementation results in minimal sodium loading or exacerbation of hypertension.[210,212]

However, these positive trials have been limited by lack of blinding and a placebo control. More recently, double-blind placebo-controlled trials were conducted, using sodium bicarbonate or veverimer (a novel intestinal hydrochloric acid binder) aiming to slow CKD progression in patients with CKD and metabolic acidosis. These higher-quality, lower-bias studies failed to show any benefit of slowing CKD progression with treatment of mild-to-moderate metabolic acidosis. Specifically, the VALOR-CKD study reported that ververimer does not slow CKD progression (defined as development of ESKD, a sustained decline in eGFR of ≥40% from baseline, or death due to kidney failure), with a hazard ratio of 0.99 (95% CI, 0.8–1.2).[212a] As such, the most recently updated KDIGO CKD guidelines recommend that patients maintain serum bicarbonate concentrations >18 mmol/L—guidance based largely on opinion rather than evidence. At present, evidence does not support treatment of metabolic acidosis to slow CKD progression.[213,214]

NOVEL BIOMARKERS

PLASMA ASYMMETRIC DIMETHYLARGININE

Asymmetric dimethylarginine (ADMA) is formed by the breakdown of arginine-methylated proteins and acts as an endogenous inhibitor of nitric oxide synthase to reduce nitric oxide production. The higher serum ADMA concentrations observed with impaired kidney function have been proposed as one mechanism for the endothelial dysfunction associated with CKD. Elevated ADMA levels are associated with CVD and cardiovascular mortality in patients with CKD.[215] In animal models, administration of ADMA resulted in the development of hypertension, increased deposition of collagen I and III and fibronectin in glomeruli and blood vessels, and rarefaction of peritubular capillaries.[216] Conversely, the overexpression of dimethylarginine dimethylaminohydrolase, the enzyme responsible for degradation of ADMA, was associated with reduced plasma ADMA concentrations and amelioration of kidney injury in rats after 5/6 nephrectomy, implying that circulating ADMA may also promote CKD progression.[217]

Among 131 patients with CKD, a higher plasma ADMA concentration was associated with ESKD or death (HR, 1.20; 95% CI, 1.07–1.35 for each 0.1 μmol/L increase).[161] In 227 relatively young patients without diabetes with mild-to-moderate CKD, higher plasma ADMA levels predicted progression to the combined endpoint of creatinine doubling or ESKD (HR, 1.47; 95% CI, 1.12–1.93 for each 0.1 μmol/L increase).[218] Finally, retrospective analysis of data from 109 patients with IgA nephropathy showed associations between plasma ADMA concentrations and glomerular and tubulointerstitial injury. Furthermore, the plasma ADMA concentration was an independent determinant of the annual rate of decline in GFR.[219]

SERUM PHOSPHATE AND FGF23

When rats were fed a high-phosphate diet after uninephrectomy, renal calcium and phosphate deposition, as well as tubulointerstitial injury, were observed within 5 weeks.[220] Furthermore, in animals and humans with CKD, dietary phosphate restriction or treatment with oral phosphate binders was associated with reductions in proteinuria and glomerulosclerosis and attenuation of CKD progression.[221-224] Together, these data suggest that phosphate loading and/or hyperphosphatemia exacerbate kidney injury in CKD. Three cohort studies of patients with CKD have identified higher serum phosphate concentrations as an independent risk factor for progression.[206,225,226] By contrast, the largest study to date, which included 10,672 participants with CKD, found no independent association between higher serum phosphate and risk of progression.[227] It should be noted, however, that the number of ESKD events was low, and the study therefore had limited power to detect a moderate association between serum phosphate and CKD progression.[228] In addition, higher serum concentrations of the phosphatonin fibroblast growth factor 23 (FGF23), a hormone involved in calcium and phosphate regulation through decreasing serum phosphate and 1,25-dihydroxyvitamin D_3, have been associated with CKD progression.[229,230] Along with adverse kidney effects, higher serum concentrations of FGF23 are associated with CVEs and related surrogate outcomes in patients with CKD.[231-233] In the CRIC study, higher serum concentrations of FGF23 were determined to be an independent risk factor for ESKD, but more so in patients with a higher baseline eGFR (HR, 1.3; 95% CI, 1.04–1.6 if the eGFR was 30–44 mL/min/1.73 m^2; and HR, 1.7; 95% CI, 1.1–2.4 if the eGFR was >45 mL/min/1.73 m^2).[231] A meta-analysis confirmed the association between elevated serum FGF23 and CVEs, as well as all-cause mortality, but suggested that the relation was not causal due to lack of an exposure-response relationship.[233] The exact clinical significance of FGF23 is uncertain, but there appears to be potential for its use as a biomarker, specifically in patients with CKD and ESKDs.[149]

NEUTROPHIL GELATINASE–ASSOCIATED LIPOCALIN

Neutrophil gelatinase–associated lipocalin (NGAL) is an iron-carrying protein expressed throughout the distal tubule epithelium and is overexpressed in response to AKI.[234] While it is known mostly for its role in AKI, the role of urinary NGAL in CKD progression is currently a point of interest.[235] The Atherosclerosis Risk in Communities (ARIC) study found that baseline urinary NGAL was highest in participants with incident CKD (OR, 2.1; 95% CI, 0.96–4.6).[236] In the CRIC study, urinary NGAL was associated with ESKD (HR, 1.7; 95% CI, 1.2–2.5) but added no significant improvement in prediction of CKD progression beyond eGFR decline and proteinuria.[237]

KIDNEY INJURY MOLECULE-1

Kidney injury molecule-1 (KIM-1) is a transmembrane glycoprotein typically undetected in a normal kidney but detected in patients with AKI and CKD.[238] One cohort study found that serum KIM-1 concentrations increased with higher CKD stage and were directly associated with eGFR decline and ESKD.[239] The Multi-Ethnic Study of Atherosclerosis determined that doubling of urinary KIM-1 concentrations was associated with CKD and rapid eGFR decline (>3 mL/min/1.73m^2 per year, OR, 1.2; 95% CI, 1.02–1.3).[240] Studies combining KIM-1 with other markers (TNFR1, TNFR2) have also shown improvement in prediction in patients with diabetic kidney disease, but larger studies examining this approach and adjusting for cystatin C-based eGFR rather than creatinine-based eGFR, are needed.[241,242]

SOLUBLE UROKINASE-TYPE PLASMINOGEN ACTIVATOR RECEPTOR

Soluble urokinase-type plasminogen activator receptor (suPAR) is a protein involved in cell adhesion and migration of endothelial and immune cells. Increased plasma suPAR concentrations have been detected in patients with glomerular disease, specifically glomerulosclerosis, as well as adverse CVEs.[149] One study done by Hayek and colleagues[243] using 2292 patients from the Emory Cardiovascular Bank found that higher baseline plasma suPAR concentrations were associated with CKD (eGFR <60 mL/min/1.73 m^2). This relation was further validated using a sample of 347 patients in the Women's Interagency Human Immunodeficiency Virus Study, where again a higher baseline plasma suPAR concentration was associated with higher incidence of CKD. Specificity may be an issue, however. For example, 30% of patients with an eGFR > 90 mL/min/1.73m^2 also had an increase in baseline plasma suPAR concentrations, indicating that suPAR may be linked to inflammation or other factors in addition to kidney function.[243] These results emphasize the need for further validation.

UROMODULIN

Uromodulin (UMOD) is a kidney-specific glycoprotein synthesized by epithelial cells found in the ascending loop of Henle. While its exact function is currently unknown, one proposed role has been in protection against urinary tract infections, innate immunity activation, and kidney stone prevention. Genome-Wide Association Studies (GWASs) indicated that single-nucleotide polymorphism (SNP) variants of the *UMOD* gene were associated with lower eGFR and development of CKD in European cohorts.[56,244] One GWAS using 3203 Icelandic patients with CKD found an association between CKD and serum creatinine with a variant adjacent to the *UMOD* gene on chromosome 16p12, which was strengthened with age. Whether UMOD is a marker of CKD progression remains unclear. A meta-analysis of GWAS for UMOD including six studies and 10,884 patients found that two loci located around the *UMOD* gene were associated with increased urinary UMOD levels.[245] However, Shlipak and colleagues[246] found that among 879 individuals of the Heart and Soul Study, UMOD SNP variants were associated with urine UMOD concentrations but did not affect incident CKD risk. These conflicting results indicate that further investigation is required before clinical implementation of UMOD as a CKD biomarker.[149]

PROTEOMIC APPROACHES

Proteomics simultaneously assess multiple proteins in blood and/or urine using electrophoresis and mass spectrophotometry in both targeted (known pathophysiology) and nontargeted (unknown pathophysiology) approaches.[247] However, this approach poses several challenges given biosampling variability due to age, sex, diet, exercise, and other issues. Using proteomics in a Scottish cohort with type II DM, investigators for the Innovative Diabetes Tools study were able to identify at least 62 of 207 serum protein biomarkers associated with rapid progression of CKD (>40% baseline eGFR loss in 3.5 years). The use of a 14-biomarker panel (including FGF23 and KIM-1) in risk prediction models allowed for an increase in the area under the receiver operating characteristic curve (AUROC) from 0.706 to 0.868.[248] Another study, which used capillary electrophoresis with mass spectrophotometry, looked at the utility of urinary proteomics in validating previously established biomarkers of CKD. The study used a general population cohort of 223 healthy individuals and 1767 patients with CKD (defined as eGFR <90 mL/min/1.73 m^2 or urine albumin excretion >30 mg/L), finding that the proteome performed better in detection and prediction of CKD progression, improving the AUROC based on baseline eGFR and albuminuria from 0.76 to 0.82.[249] A large proteomic study assessed 4638 plasma proteins in 3235 participants from CRIC using 50% decline in eGFR or kidney failure over 10 years as the primary outcome and validated the findings in ARIC. One hundred proteins were associated with the outcome and risk model that included 65 proteins and produced good discrimination (C-statistic [95% CI] = 0.86 [0.84, 0.89]).[250] The use of novel serum or urinary biomarkers will likely be of value in future risk prediction models through enhancing discrimination beyond traditional risk predictors including GFR and proteinuria. Given the interplaying mechanisms that may result in any given patient with CKD, it is unlikely that any one single conventional or novel biomarker will have a large effect on risk prediction models. However, the development of biomarker panels through the use of proteomics shows potential in discriminating between disease and nondisease states, as well as risk prediction, and its future role in the clinical setting, is promising.[149]

OTHER BIOMARKERS

A number of other biomarkers are currently being investigated as risk factors in CKD. Although many have been reported to be associated with adverse outcomes, the challenge is to identify biomarkers that add to the predictive power of established risk factors.

ENVIRONMENTAL RISK FACTORS

SMOKING

Population-based studies have identified cigarette smoking as an independent risk factor for various manifestations of CKD including proteinuria,[251] elevated serum creatinine concentrations,[252] decreased eGFR,[23,253] and development of ESKD or death associated with CKD (HR, 2.6; 95% CI, 1.8–3.7).[30] In the latter study, 31% of the attributable risk of CKD was associated with smoking. In a longitudinal study of 10,118 middle-aged Japanese workers, smoking was associated with an increased risk of developing glomerular hyperfiltration (eGFR ≥117 mL/min/1.73 m^2; OR, 1.32 vs. nonsmokers) and proteinuria (OR, 1.51 vs. nonsmokers).[254] Two other similar longitudinal studies from Japan have confirmed that smoking is associated with an increased risk of developing proteinuria but a higher mean eGFR than in nonsmokers.[255,256] In one study, smoking was associated with a lower risk of developing CKD stage 3.[256] Smoking has been shown to increase the risk of progression of CKD due to diabetes,[257,258] hypertensive nephropathy,[259] glomerulonephritis,[260] lupus nephritis,[261] IgA nephropathy,[262] and adult PKD.[262] Randomized trials of the effect of smoking cessation on CKD progression are lacking but, in one observational study in patients with diabetes, smoking cessation was associated with less frequent progression to macroalbuminuria and a slower rate of GFR decline than continued smoking.[263] Similarly, in CRIC, nonsmoking was associated with a reduced risk of CKD progression (HR, 0.68; 95% CI, 0.55–0.84), atherosclerotic CVEs (HR, 0.55; 95% CI, 0.40–0.75), and mortality (HR, 0.45; 95% CI, 0.34–0.60).[264] Among 9270 participants with CKD in the SHARP, current smoking was associated with an increased risk of all-cause mortality, CVEs, and cancer but was unassociated with risk of ESRD or rate of GFR decline in 6245 participants not receiving dialysis at baseline.[265] Possible mechanisms whereby cigarette smoking may contribute to kidney damage include sympathetic nervous system activation, glomerular capillary hypertension, endothelial cell injury, and direct tubulotoxocity.[266]

ALCOHOL

The role of alcohol consumption as a potential risk factor for CKD remains unclear. One case-control study found a significant association between ESKD and consumption of more than two alcoholic drinks daily,[267] whereas another similar study found no association (with the exception of "moonshine").[268] Some population-based studies have found that alcohol consumption is not related to CKD risk,[269-271] but one study found a significant association of heavy alcohol intake (more than four drinks daily) and prevalent CKD, as well as the risk of developing CKD in participants with a normal GFR.[253] Furthermore, heavy alcohol intake substantially increased the risk of CKD progression associated with smoking, such that participants who smoked and drank heavily had an almost fivefold increased risk of developing CKD.[253]

Conversely, several large cohort studies have reported an inverse relation between alcohol consumption and the risk of developing CKD[272,273] or ESKD.[274] Another study found that moderate-to-heavy alcohol consumption was associated with a higher risk of developing albuminuria but a lower risk of eGFR <60 mL/min/1.73m^2.[275] This finding of a lower risk of eGFR decline is likely due to sarcopenia associated with heavy alcohol use rather than a true improvement in kidney function.

The most rigorous study published to date investigated the incidence of CKD defined by an eGFR determined using the combined cystatin C and creatinine equation or albuminuria >30 mg/day based on two consecutive 24-hour urine collections. The risk of developing CKD over a mean of 10.2 years decreased progressively with increasing alcohol consumption: HR of 0.85 (95% CI, 0.69–1.04) for occasional alcohol consumption (<10 g/week), HR of 0.82 (95% CI, 0.69–0.98) for light alcohol consumption (10–69.9 g/week), HR of 0.71 (95% CI, 0.58–0.88) for moderate alcohol consumption (70–210 g/week), and HR of 0.60 (95% CI, 0.42–0.86) for heavier alcohol consumption (>210 g/week).[276]

RECREATIONAL DRUGS

The role of recreational drugs as a risk factor for CKD has not been widely studied, but one case-control study reported a positive association among heroin, cocaine, or psychedelic drug use and ESKD.[277] Following reports of a specific renal lesion characterized by proteinuria and FSGS, termed "heroin nephropathy," other investigators reported a wide range of inpatients with a history of heroin abuse, confirmed by biopsy. It is unclear whether the observed kidney lesions resulted from direct effects of heroin or were attributable to impurities in the drug or associated blood-borne virus infections and endocarditis. An association with renal amyloidosis, possibly due to chronic skin infections, has also been reported.[278] Interestingly, heroin abuse was not associated with an increased risk of mild CKD in 647 hypertensive patients who showed an association between illicit drug abuse and CKD.[279] Cocaine exerts several adverse effects that may induce kidney injury including rhabdomyolysis, vasoconstriction, activation of the renin-angiotensin–aldosterone system, oxidative stress, and increased collagen synthesis.[278] Furthermore, chronic administration of cocaine to rats resulted in multiple renal lesions including glomerular atrophy and sclerosis, tubule cell necrosis, and areas of interstitial necrosis.[280] Among 647 patients attending a hypertension clinic, a history of any illicit drug use was independently associated with a relative risk of 2.3 (95% CI, 1.0–5.1) for mild CKD, whereas cocaine and psychedelic drug use were associated with relative risks of 3.0 (95% CI, 1.1–8.0) and 3.9 (95% CI, 1.1–14.4), respectively.[279] In one prospective cohort study using 2286 participants of the Life Span study, use of opiates and cocaine were both determined to be associated with eGFR <60 mL/min/1.73 m^2; OR, 2.71 and 95% CI, 1.50–4.89 for opiates; OR, 1.40 and 95% CI, 0.87–2.24 for cocaine); cocaine use was associated with a higher odds of albuminuria (UACR >30 mg/g; OR, 1.80 and 95% CI, 1.29–2.51.[281]

ANALGESICS

Analgesic nephropathy has been well described as a cause of CKD and ESKD resulting from abuse of combination analgesics containing aspirin and phenacetin prevalent in Australia and Switzerland until the sale of these products was restricted.[282] Cohort studies of participants without CKD at baseline have not consistently identified associations between analgesic use and incident CKD. Among 1697 women in the Nurses Health Study, consumption of >3000 g of acetaminophen was associated with an increased risk of eGFR decline >30 mL/min/1.73 m^2 over 11 years (HR, 2.04; 95% CI, 1.28–3.24); use of aspirin or nonsteroidal anti-inflammatory drugs (NSAIDs) was not associated with increased risk.[283] Among 4494 male physicians from the Physicians Health Study, there was no association between occasional-to-moderate use of aspirin, acetaminophen, or NSAIDs and GFR decline over 14 years.[284] By contrast, analgesic use may exacerbate the progression of established CKD. In one large study of 19,163 patients with newly diagnosed CKD, use of aspirin, acetaminophen, or NSAIDs was associated with an increased risk of progression to ESKD in a dose-dependent manner. Among cyclooxygenase 2 inhibitors, use of rofecoxib but not celecoxib was associated with increased risk of ESKD.[285] In a cohort study of 4101 people with rheumatoid arthritis, chronic use of NSAIDs was not associated with a more rapid GFR decline in the entire study population, but NSAID use was independently associated with more rapid GFR decline in a small minority of patients with CKD stage 4 or 5 (N = 17).[286] A meta-analysis of three studies that included data from 54,663 participants with CKD stages 3 to 5 found no association between regular NSAID use and accelerated GFR decline (defined as ≥15 mL/min over 2 years) but did report an association with high-dose NSAID use (defined as 90th percentile or above in one study and not defined in the other) in two studies.[287] The use of single-compound acetaminophen or aspirin was reported not to accelerate progression among patients with CKD stage 4 or 5,[288] but a systematic review of the safety of acetaminophen treatment reported an increased risk of renal adverse events in three of four observational studies, in addition to an increased risk of all-cause mortality and cardiovascular and gastrointestinal adverse events in other studies.[289]

As the authors noted, these results must be interpreted with caution due to the possibility of confounding by indication resulting from associations between the indication for prescribing selected analgesic agents and adverse outcomes.

HEAT STRESS

In more tropic parts of the globe including Latin America, Sri Lanka, and India, chronic kidney disease of unknown etiology (CKDu) has been reported. Previously known as Mesoamerican nephropathy given its initial identification in Central American farmworkers, CKDu pathology is thought to be related to heat-induced dehydration. Other factors including socioeconomic status, exposure to pesticides through farm labor, and poor-quality drinking water have been considered. For a more detailed discussion of this topic, see Chapter 83.[290-292,293,294]

HEAVY METALS

Chronic exposure and subsequent toxicity of heavy metals, specifically lead and cadmium, has been known to result in a predominantly tubulointerstitial nephropathy. Overt lead toxicity is characterized by chronic interstitial nephritis and an association with gout. In addition, epidemiologic studies have reported that mild elevations in blood lead concentrations are associated with moderate reductions in GFR and/or hypertension in the general population.[295,296] Furthermore, a prospective study identified elevations in blood lead concentrations and body lead burden even within the population reference range as risk factors for CKD.[297] Similarly, blood lead burden was a risk factor for progression among 108 patients with low-normal values and no history of lead exposure.[298] Furthermore, randomization to chelation therapy was associated with a modest improvement in GFR over 24 months versus a small decline in those randomized to control (+6.6 ± 10.7 vs. −4.6 ± 4.3 mL/min/1.73 m^2; P < .001).[298]

Like lead, chronic exposure to cadmium is also associated with a distinctive nephropathy, characterized by proximal tubule damage and low-molecular-weight proteinuria.[299] Furthermore, low-level cadmium exposure resulting from environmental contamination was associated with tubular proteinuria[300]; and analysis of data from 14,778 participants in NHANES showed an independent increased risk of albuminuria, reduced GFR, or both when comparing the highest and lowest quartiles of blood cadmium concentrations.[301] Comparison of the lowest and highest quartiles for whole blood concentrations of cadmium and lead showed an even greater increased risk of albuminuria, reduced GFR, or both.[301] In another NHANES study, blood and urine cadmium levels were directly correlated with UACR and inversely correlated with GFR. Higher blood and urine cadmium concentrations were independently associated with albuminuria, and higher blood cadmium concentrations were associated with albuminuria and a reduced GFR.[299] Occupational or low-level environmental exposure to cadmium was associated with an increased risk of ESKD in a population-based study from Sweden.[302]

RENAL RISK SCORES

The focus on investigating risk factors that predict the development and/or progression of CKD in diverse populations has led to the observation that a relatively small group of risk factors appears to be common in different forms of CKD. This supports the notion of a common pathway of mechanisms that underlie the progression of CKD. It has also led to the proposal that these common risk factors could be combined to develop a renal risk score to predict the development and future risk of progression of CKD in a manner analogous to the Framingham risk score for predicting cardiovascular risk in the general population.[303] The revised classification system for CKD proposed by KDIGO has in part addressed this need by incorporating the evidence that a reduced GFR and albuminuria are both powerful risk factors to incorporate into a system in which CKD categories correspond to risk of progressive CKD.[10]

In addition, considerable progress has been made in developing renal risk scores to facilitate more accurate risk prediction. These may conveniently be divided into two groups—those that apply to the general population (i.e., without CKD as baseline) and those that predict the risk of progression in patients already diagnosed with CKD. In addition, one study has developed a risk score to predict the development of CKD after an episode of AKI.

METHODS OF KIDNEY RISK SCORE DEVELOPMENT

Before clinical implementation, renal risk scores must first hold both internal and external validity, show improvement over the current risk classification system, and be easily integrated into clinical settings. The presence of internal validity indicates that the prediction model has been developed from a sample that accurately reflects the relations among the variables used in the prediction model and outcome of interest. These predictor variables must have clear definitions and be measured in each patient well before the final outcome is reached. The outcome itself, in order to avoid bias, must also be clearly defined and assessed uniformly in all patients, blind to the status of the predictor variable.[206] In order to develop such a risk model, appropriate statistical approaches must be taken. If censoring of the variable is negligible and the follow-up period is clearly defined, logistic regression can be used. However, if there is significant censoring present, Cox proportional hazards regression is preferred.[304,305] Competing risk models can also be considered when there is a concomitant high risk of mortality and kidney failure in the same population, but conclusive evidence to support the routine use of a competing risk framework for CKD progression is insufficient.[306] Finally, machine learning approaches may have some advantages for predicting progression, particularly with handling of missing data, but they should only be considered in large datasets with external validation due to concerns about overfitting.[241,307,308] In addition to the general validation issues discussed earlier, several metrics specific to performance of the prediction model are used in order to assess internal validity.

METRICS OF MODEL PERFORMANCE

DISCRIMINATION

Discrimination in this context is generally defined as a model's ability to accurately assign higher probabilities to patients in whom the event of interest occurs, compared with those in whom it does not. The concordance or C-statistic, which is the most commonly used tool of discrimination, is defined as the proportion of times the prediction model correctly discriminates between a randomly selected pair of individuals (case and control) and can be considered identical to the area under the receiver operator curve (AUROC). Like the AUROC, a C-statistic of 0.50 indicates that the model performance is no better than chance; a C-statistic of 0.70 to 0.80 indicates that the model has a good performance in discrimination; and a C-statistic of >0.80 generally indicates excellent performance in discrimination. The comparison of C-statistics is a frequently used method for comparing the discrimination of multiple prediction models in order to determine which one is superior in risk prediction.[309-312] However, as the value of the C-statistic approaches 1, it becomes difficult to appreciate a significant difference in the values of different models and a more sensitive method, such as the Integrated Discrimination Index, should be considered. The Integrated Discrimination Index looks at the difference in the discrimination slopes of the two separate models and describes it on an absolute and relative scale, achieving an effective comparison of discrimination values when the C-statistics are no longer sufficient.[313,314]

CALIBRATION

Model calibration is another metric of model performance, referring to how well the predictions made from the model align with actual data. For logistic regression models, the Hosmer–Lemeshow chi-square statistic is the most commonly used method for such assessments. This method ranks participants based on predicted probability into deciles, and the mean probability of each decile is then compared with the actual frequency of outcomes among participants in each decile. A chi square is then used to assess for significant discrepancies between the predicted and actual probabilities. Chi-square statistics that are said to be significant indicate that the model calibration is suboptimal.[315] The metric of calibration is valuable for clinical risk prediction models because a poorly calibrated model will result in either underprediction or overprediction of risk.[206]

RECLASSIFICATION

It is standard for clinical treatments and tests to be selected on the basis of the predicted risk of having an event. In the development of a new prediction model, it is essential to consider whether it reclassifies patients into more appropriate risk categories than was the case in the older or standard prediction model. If a patient has an event and the new model has assigned the said patient to an appropriately higher-risk category, the new model will be considered a success. Similarly, for patients who do not have an event, the new model is successful if it reclassifies the patient to a lower-risk category than the previous model. However, if the new prediction model reassigns the patient to a risk category that is opposite of the actual outcome, the new model is considered unsuccessful and is not superior (or potentially inferior) to the previous model. This success or failure can be quantified by the Net Reclassification Index, values for which range from –2.0 to +2.0, with positive values indicating successful reclassification and negative values indicating unsuccessful reclassification.[313,314]

CLINICAL UTILITY

Even the best risk prediction model is unusable in a clinical setting if it cannot be rapidly and efficiently implemented. As previously indicated, prediction models are derived through a series of complex analytics models that cannot be easily applied to patients using simple calculations. This means that the model, once validated, needs to be translated into a simpler clinically useful bedside tool. However, in simplifying the scoring system, there is some inevitable loss of precision, discrimination, and calibration. The advent of rapid access to Web-based calculators or smartphone apps has allowed for complex prediction models to be applied in their original form via a simplified user interface and without the need for complex calculations by the user.[316-319]

EXTERNAL VALIDITY

External validity, unlike internal validity, addresses whether the results of the study sample can be applied to the general population. The external validity of a prediction model can never be assumed because it is likely that a model generated from one set of data (derivation cohort) will not perform exactly the same in other cohorts. This is likely due to underlying biological factors including differences in disease or physiology of the cohorts and false associations between predictors and outcomes in the original derivation cohort. Attempts to minimize errors between cohorts can be achieved by careful selection of cohorts that are representative of the clinical condition and by choosing predictor variables based on clinical relevance rather than statistical associations.[312]

MODELS PREDICTING INCIDENT CHRONIC KIDNEY DISEASE

Risk scores have been proposed to assess the risk of developing CKD in the general population and, in some cases, its subsequent progression. These are summarized in Table 19.3 and have been assessed in a systematic review.[320]

In one study, data from 8530 adults included in NHANES were used to identify risk factors for prevalent CKD (defined as eGFR <60 mL/min/1.73 m^2). The authors proposed a risk score that included age, female sex, hypertension, anemia, diabetes, peripheral vascular disease, history of CVD,

Table 19.3 Kidney Risk Scores for the General Population

	Study					
Parameter	**SCORED**[321]	**SCORED2**[351]	**Chinese**[352]	**Framingham**[353]	**QKIDNEY**[322]	**PREVEND**[354]
Population	NHANES	CHS + ARIC	General population	FHS	QResearch	eGFR >45
Outcome	eGFR <60 mL/min/1.73 m^2 (prevalent)	eGFR <60 mL/min/1.73 m^2 (incident)	eGFR <60 mL/min/1.73 m2 (incident)	eGFR <60 mL/min/1.73 m2 (incident)	CKD, ESKD	Rapid ↓ GFR
Factor	Age	Age	Age	Age	Age	Age
	Female	Female				
					Ethnicity	
					Deprivation	
					Family history	
					Smoking	
			BMI			
	HT	HT		HT	HT	HT
	DM	DM	Type 2 DM	DM	DM	
	PVD	PVD			PVD	
	CVD	CVD	Stroke		CVD	
	CCF	CCF			CCF	
					RA	
	Anemia	Anemia				
			DBP		SBP	SBP
					BMI	
				eGFR		eGFR
	Proteinuria		Proteinuria	Albuminuria		Albuminuria
						CRP
			Uric acid			
			HbA1c			
			Glucose			
					NSAIDs	
AUC	0.88	0.69	0.77	0.81	0.88	0.84
Validation	ARIC	CHS + ARIC	General population	ARIC	THIN	Internal

ARIC, Atherosclerosis Risk in Communities; *AUC,* area under the receiver operator curve; *BMI,* body mass index; *CCF,* congestive cardiac failure; *CHS,* Cardiovascular Health Study; *CKD,* chronic kidney disease; *CRP,* C-reactive protein; *CVD,* cardiovascular disease; *DBP,* diastolic blood pressure; *DM,* diabetes mellitus; *eGFR,* estimated glomerular filtration rate; *ESKD,* end-stage kidney disease; *FHS,* Framingham Heart Study; *HbA$_{1c}$,* hemoglobin A$_{1c}$; *HT,* hypertension; *NHANES,* National Health and Nutrition Examination Survey; *NSAIDs,* nonsteroidal anti-inflammatory drugs; *PREVEND,* Prevention of Renal and Vascular End-stage Disease; *PVD,* peripheral vascular disease; *RA,* rheumatoid arthritis; *SBP,* systolic blood pressure; *THIN,* the Health Improvement Network.

congestive heart failure, and proteinuria. The AUROC was high at 0.88, and a score of ≥4 resulted in a sensitivity of 92% and specificity of 68%. The positive predictive value was low at 18%, but the negative predictive value was 99%. External validation using data from the ARIC study gave an AUROC value of 0.71.[321] This was a cross-sectional study, and the risk score therefore did not predict the risk of future CKD. Rather, it identified individuals at increased risk of having current undiagnosed CKD. As such, it might be useful for guiding efforts to screen populations for CKD but gives no information about the future risk of CKD progression. The applicability of the score to general populations is somewhat weakened by the inclusion of two variables that require prior laboratory testing: anemia and proteinuria. Furthermore, the presence of significant proteinuria is sufficient to diagnose CKD in the absence of any reduction in GFR.

Most scores in Table 19.3 are useful to identify individuals at higher risk of developing CKD for monitoring or intervention to reduce risk, but they do not distinguish the minority who are at risk of progressing to ESKD from the majority who are at low risk. To identify only high-risk individuals, another group used data from 775,091 women and 799,658 men aged 35 to 74 years, without a recorded diagnosis of CKD in 368 primary care practices in the United Kingdom to develop a risk score. Two outcomes were studied over a period of up to 7 years—moderate-to-severe CKD (defined as kidney transplantation, dialysis, diagnosis of nephropathy, proteinuria, or eGFR <45 mL/min/1.73 m^2) and ESKD (defined as kidney transplantation, dialysis, or eGFR <15 mL/min/1.73 m^2); separate risk scores were developed for men and women. The final model for moderate-to-severe CKD included age, ethnicity, social deprivation, smoking, BMI, systolic blood pressure, diabetes, rheumatoid arthritis, CVD, treated hypertension, congestive cardiac failure, peripheral vascular disease, use of NSAIDs, and family history of kidney disease. In women, it also included systemic lupus erythematosus and history of kidney stones. The model for ESKD was similar but did not include NSAID use. Internal and external validation was performed, giving AUROC values of 0.818 to 0.878.[322] This study also illustrates the utility of a risk score that could be programmed into primary care computer systems to alert family practitioners to patients who are at risk of progression to ESKD.

MODELS PREDICTING KIDNEY FAILURE

Several risk scores have been developed for patients with diagnosed CKD in a variety of study populations and are summarized in Table 19.4 and in systematic reviews.[320,323]

Analysis of data from 1513 patients with diabetic nephropathy included in the RENAAL study identified UACR, serum albumin, serum creatinine, and hemoglobin as independent risk factors for ESKD. A risk score was

Table 19.4 Kidney Risk Scores for Patients With Chronic Kidney Disease (CKD)

	Study					
Parameter	**RENAAL[164]**	**AIPRD[324]**	**IGAN[169]**	**KPC[325]**	**CRIB[326]**	**KFRE[206]**
Disease studied	DN	CKD	IgAN	CKD, stage 3 or 4	CKD, stages 3 to 5	CKD, stages 3 to 5
Variables		Age	Age	Age		Age
			Male	Male	Female	Male
	Creatinine	Creatinine	1/creatinine	eGFR	Creatinine	eGFR
	UACR	UPE	Proteinuria	N/A	UACR	UACR
		SBP	SBP	HT		
				DM		
	Alb		TP			Alb
	Hb			Anemia		
						Calcium
					Phos	Phos
						Bicarb
			Histol			
			Hematuria			
Outcome	ESKD	ESKD or doubling of serum creatinine level	ESKD	RRT	ESKD	ESKD
AUC			0.939	0.89	0.873	0.917
Validation	No	No	No	No	Yes	Yes

AIPRD, ACE Inhibition in Progressive Renal Disease study; *Alb,* serum albumin; *AUC,* area under the receiver operator curve; *Bicarb,* serum bicarbonate; *CRIB,* Chronic Renal Impairment in Birmingham study; *DM,* diabetes mellitus; *DN,* diabetic nephropathy; *eGFR,* estimated glomerular filtration rate; *ESKD,* end-stage kidney disease; *Hb,* hemoglobin; *Histol,* histologic grade; *HT,* hypertension; *IgAN,* IgA nephropathy; *KPC,* Kaiser Permanente Cohort; *N/A,* not available; *Phos,* serum phosphate; *Proteinuria,* urine dipstick proteinuria; *RENAAL,* Reduction of Endpoints in NIDDM with the Angiotensin II Antagonist Losartan study; *RRT,* renal replacement therapy; *SBP,* systolic blood pressure; *SHC,* Sunnybrook Hospital cohort; *TP,* serum total protein; *UACR,* urine albumin-to-creatinine ratio; *UPE,* 24-hour urinary protein excretion.

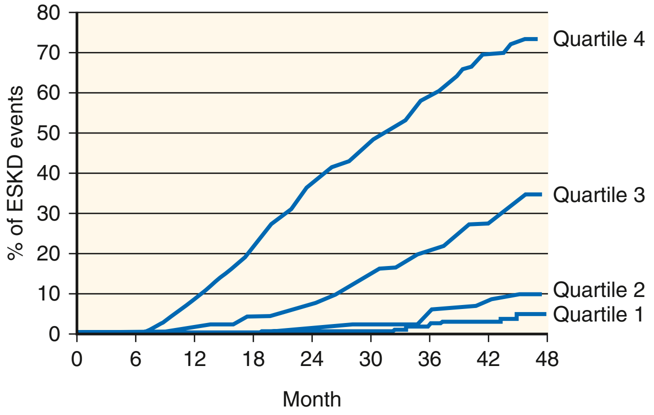

Fig. 19.5 Kaplan-Meier curve for the end-stage kidney disease (ESKD) endpoint stratified by quartile of risk score in 1513 patients with diabetic nephropathy from the Reduction of Endpoints in NIDDM with the Angiotensin II Antagonist Losartan (RENAAL) study. *NIDDM,* Non–insulin-dependent diabetes mellitus. (From Keane WF, Zhang Z, Lyle PA, et al. Risk scores for predicting outcomes in patients with type 2 diabetes and nephropathy: the RENAAL study. *Clin J Am Soc Nephrol*. 2006;1:761–767.)

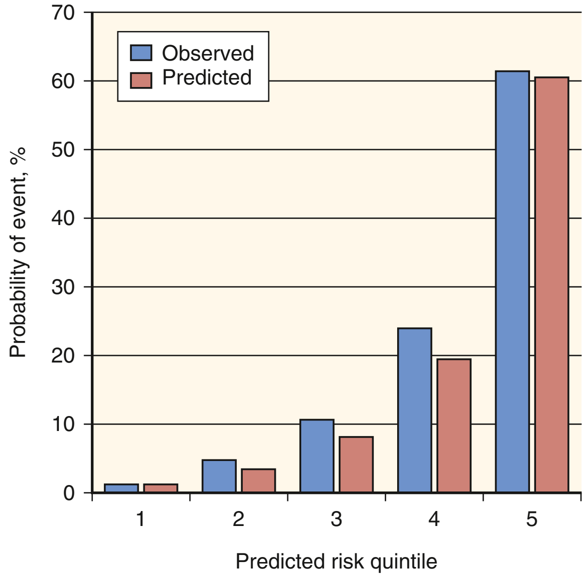

Fig. 19.6 Renal risk score predicted versus observed risk of developing end-stage kidney disease at 3 years in a validation cohort of patients with chronic kidney disease stages 3 to 5. (From Tangri N, Stevens LA, Griffith J, et al. A predictive model for progression of chronic kidney disease to kidney failure. *JAMA.* 2011;305:1553–1559.)

derived from the coefficients of these variables in the Cox proportional hazards model, which successfully separated the participants into quartiles of ESKD risk (Fig. 19.5), with a marked difference in risk between the first and last quartiles (6.7 vs. 257.2/1000 patient years).[164]

Among 1860 patients with nondiabetic CKD from a combined database of 11 clinical trials, Cox proportional hazards regression analysis identified age, serum creatinine, proteinuria, and systolic blood pressure as independent risk factors for the combined endpoint of time to ESKD or creatinine doubling. Using similar methodology as the previous study, a risk model based on these variables was developed to stratify patients into quartiles of risk. The annual incidence of the combined endpoint was 0.4% versus 28.7% in the lowest versus highest quartile for patients in the control group and 0.2% versus 19.7% in those randomized to angiotensin-converting enzyme inhibitor treatment.[324] Analysis of data from 2269 patients with IgA nephropathy identified systolic blood pressure, proteinuria (assessed with urine dipstick test), serum total protein, 1/serum creatinine, and histologic grade at initial biopsy as predictors of time to ESKD Age, gender, and severity of hematuria were added to these variables to develop a scoring system for estimating 4- and 7-year cumulative incidence of ESKD. There was close agreement between estimated and observed risks (AUROC value = 0.939).[169]

Another study evaluated 9782 patients with CKD stages G3 and G4, aiming to predict the onset of kidney replacement therapy (dialysis or transplantation). Six independent risk factors were identified—age, male sex, eGFR, hypertension, diabetes, and anemia—and were incorporated into a risk score that stratified participants into quintiles of risk. The risk of progression to RRT was 19% in the highest risk quintile versus 0.2% in the lowest. The AUROC value was 0.89, and observed risk differed from predicted by <1%.[325] One limitation of this study was the lack of data on proteinuria or albuminuria.

Among 382 patients with CKD stages 3 to 5 from the Chronic Renal Impairment in Birmingham (CRIB) study, independent risk factors for progression to ESKD during a mean of 4.1 years' follow-up were female sex, serum creatinine, serum phosphate, and UACR. A risk score was derived from these variables (AUROC value = 0.873) and externally validated in a similar cohort of patients with CKD stages 3 to 5 (East Kent cohort), giving an AUROC value of 0.91, even though the urine ACR was not available in the validation cohort.[225]

KIDNEY FAILURE RISK EQUATIONS

The kidney failure risk equations (KFREs) for predicting ESKD were developed by Tangri and colleagues[206] using data from a cohort of 3449 Canadian patients with CKD stages 3 to 5 (Sunnybrook Hospital cohort). The eight-variable equation included age, male sex, eGFR, albuminuria, serum calcium, serum phosphate, serum bicarbonate, and serum albumin (AUROC value = 0.917). External validation was performed with data from a separate cohort of 4942 patients with CKD stages 3 to 5 (British Columbia CKD Registry), giving an AUROC value of 0.841. There was close agreement between predicted and observed risk in the validation cohort (Fig. 19.6). Simpler three-variable (age, sex, and eGFR) and four-variable models (age, sex, eGFR, and albuminuria) also performed well (C-statistic [equivalent to AUROC] of 0.89 and 0.91, respectively, in the development cohort; Table 19.5), but the eight-variable equation (C-statistics 0.92) performed better in the validation cohort with respect to discrimination (C-statistic, 0.79 and 0.83, respectively, for the three- and four-variable equations), calibration, and reclassification. The authors facilitated clinical application of this risk score by producing an electronic risk calculator and a smartphone app (available for no charge at www.qxmd.com/calculate-online/nephrology/kidney-failure-risk-equation) that report an estimated risk of ESKD at 2 and 5 years. Further external validation of this risk equation has been reported by independent investigators

Table 19.5 Hazard Ratios and Goodness of Fit for Sequential Models in the Development Data Set for Three-, Four-, and Eight-Variable Kidney Failure Risk Equations

Variable	Three-Variable Model	Four-Variable Model	Eight-Variable Model
Hazard Ratio for Kidney Failure (Censored for Mortality)			
Baseline GFR, per 5 mL/min/1.73 m^2	0.54	0.57	0.61
Age, per 10 year	0.75	0.8	0.82
Male sex	1.46	1.26	1.16
Log spot urine ACR[a]		1.60	1.42
Serum albumin, per 0.5 g/dL			0.84
Serum phosphate, per 1.0 mg/dL			1.27
Serum bicarbonate, per 1.0 mEq/L			0.92
Serum calcium, per mg/dL			0.81
Goodness-of-Fit Metrics			
C-statistics	0.89	0.91	0.92
Akaike information criterion[b]	4834	4520	4432
P value[c]	<.001	<.001	<.001

[a]Hazard ratio for ACR represents a 1.0 point higher ACR on the natural log scale. For the average patient with 20 mg/g of albuminuria, this represents an increase to 55 mg/g.
[b]Null values for C-statistic and Akaike information criterion are 0.50 and 5569, respectively. Higher values for C-statistic and lower values for Akaike information criterion indicate better models.
[c]*P* values are for comparison of C-statistics between sequential models.
ACR, Albumin-to-creatinine ratio; *GFR,* glomerular filtration rate.
From Tangri N, Stevens LA, Griffith J, et al. A predictive model for progression of chronic kidney disease to kidney failure. *JAMA*.

from the MASTERPLAN (Multifactorial Approach and Superior Treatment Efficacy in Renal Patients with the Aid of Nurse practitioners) study, a cohort of 595 people from the Netherlands with CKD stages 3 to 5. The model again performed well with the eight-variable equation yielding an AUROC value of 0.89.[326] A further advantage of the Tangri risk score is that it includes variables that are all available to a biochemistry laboratory, making it possible for laboratories to automate reporting of a risk score with each eGFR.

To determine if KFREs could be implemented clinically on a global scale, a multinational validation study of the equations was performed in 35 cohorts consisting of 720,000 patients with CKD (stages 3–5) from 23 countries across four continents.[327] After a 4-year follow-up, 23,829 cases of kidney failure were observed in the cohorts. The KFRE was able to achieve excellent discrimination through all cohorts (C-statistic, 0.90; 95% CI, 0.89–0.92 at 2 years; C-statistic, 0.88; 95% CI, 0.86–0.90 at 5 years). Calibration was adequate in North American cohorts, but the original risk equations overestimated risk in some non–North American cohorts. However, through use of a simple calibration factor for non–North American populations, which lowered the baseline risk by 32.9% at 2 years and 16.5% at 5 years, calibration was improved in 12 of 15 and 10 of 13 non–North American cohorts at 2 and 5 years, respectively (P = .04 and P = .02). This indicated that the KFREs were consistently accurate in predicting progression to ESKD throughout each of the 35 cohorts with discrimination being independent of age, sex, race, or diabetic status. Moreover, the simplified four-variable equation produced similar discrimination to the eight-variable equation (at 2 years C-statistics were 0.89 vs. 0.90 and at 5 years 0.86 vs. 0.88, respectively). An online risk calculator has been produced that utilizes the four-variable equation and incorporates the correction for persons outside of North America (http://kidneyfailurerisk.com/). These findings indicated that KFREs were ready for clinical implementation.

In several countries, the KFREs are recommended for clinical use and are now included as 1A recommendations in the National Institute for Health and Care Excellence CKD guidance and KDIGO CKD guidelines. The evidence supporting their inclusion is based on studies that demonstrate improved utility in determining need for nephrology referral and vascular access planning compared with eGFR-based criteria, as well as widespread validation in more than 30 countries and 2 million individuals.[327,206,328,306]

CLINICAL RELEVANCE

KIDNEY FAILURE RISK EQUATIONS

Patients with chronic kidney disease (CKD) are at risk of kidney failure, CVEs, and early mortality. In moderate-to-severe CKD, estimated glomerular filtration rate (eGFR) and albuminuria are important functional, diagnostic, and prognostic biomarkers and can be combined in risk equations to estimate the risk of adverse events. The KFREs have been developed using eGFR, albuminuria, and readily available demographic and laboratory variables and predict the risk of kidney failure requiring dialysis or transplant with accuracy. The equations can be used to determine need for nephrology referral, intensity of care, timing of dialysis education, access formation, and preemptive transplantation. Studies evaluating the use of the risk equation in clinical settings are ongoing.[329,330]

MODELS PREDICTING DEATH AND CARDIOVASCULAR DISEASE

As mentioned before, patients with CKD progression also evidence an increased risk of death and CVD as they carry a burden of both traditional and nontraditional risk factors for CVD. Existing risk scores have shown poor prediction model metrics in patients with CKD, with no successful attempts in development of a clinically useful model for determining risk of CVD.[331] However, one prediction model developed by Bansal and colleagues[332] showed promise in assessing 5-year risk of mortality. The risk prediction model, which was developed in a cohort of 828 older adults (mean age 80 ± 5.6 years) from the CHS, included the predictor variables of age, sex, designated race, eGFR, UACR, smoking, diabetes, and history of heart failure and stroke (C-statistic = 0.72; 95% CI, 0.68–0.74). The model was externally validated in 789 participants from the Health, Aging, and Body Composition Study (C-statistic = 0.69; 95% CI, 0.64–0.74). However, patients with advanced CKD (stages 4–5) in the CHS were underrepresented, and hence the findings would require further validation before clinical implementation.[332]

A further risk model (termed KDPredict) to simultaneously predict the risk of kidney failure (defined as onset of kidney replacement therapy or sustained eGFR below 10 mL/min/1.73 m^2) and all-cause death in people with newly detected CKD category G3b-4 (eGFR 44-15 mL/min/1.73 m^2) has been published.[333] The investigators used a “super-learner” meta-algorithm to select the best prediction models that included age, sex, eGFR, albuminuria, diabetes status, and history of CVD. The models were trained on population health data from 67,942 people in Canada and validated on similar datasets from 17,528 and 7,740 people in Denmark and Scotland, respectively. The observed incidence of kidney failure was 0.8 to 1.1 per 100 patient years, whereas the incidence of death was approximately 10-fold higher at 10 to 12 per 100 patient years. Using an index of prediction accuracy based on the Brier score (which incorporates measures of both discrimination and calibration), KDPredict was reported to be more accurate than the KFRE. Importantly, KDPredict allows simultaneous assessment of the competing risks of kidney failure and death, which is helpful to inform discussions with patients regarding planning for kidney replacement therapy. One limitation is that the study population did not include persons with CKD G3a, who make up a large proportion of those with CKD. In addition, the accuracy for predicting all-cause mortality was suboptimal (AUC <0.75). Finally, the KDPredict equation also requires additional validation in populations outside of North America and Europe.

MODELS PREDICTING CHRONIC KIDNEY DISEASE AFTER ACUTE KIDNEY INJURY

A growing appreciation of the risk of CKD and progression to ESKD following an episode of AKI has prompted efforts to develop a risk score for identifying those at highest risk. Three risk prediction models were developed in one study population of 5351 predominantly male veterans with a primary admission diagnosis of AKI to predict the risk of progression to CKD stage 4. Risk factors that entered the models included increased age, low serum albumin, diabetes, lower baseline eGFR, higher mean serum creatinine levels during hospitalization, and severity of AKI as assessed by the RIFLE score (**r**isk, **i**njury, **f**ailure, **l**oss, and **e**nd-stage kidney disease) or need for dialysis, non–African American race, and time at risk. The AUROC value was 0.77 to 0.82 for the three models. Sensitivity at the optimal cut-point was 0.71 to 0.77, and specificity was 0.64 to 0.74. External validation was performed on a control population of 11,589 patients admitted for pneumonia or MI and yielded good prediction accuracy (AUROC value = 0.81 to 0.82). Sensitivity at the optimal cut-point was 0.66 to 0.71, and specificity was 0.61 to 0.70.[334]

Further validation in other populations that include a more representative proportion of women is required before these risk models can be applied. A validated risk score for patients recovering from AKI may prove important for identifying high-risk patients for closer follow-up and interventions to reduce the risk of progressive CKD, although further trials are required to evaluate the impact of renoprotective interventions in this setting.

CLINICAL IMPLEMENTATION OF RISK MODELS

Clinical implementation of risk-based care requires the interaction of both a validated risk prediction model and absolute risk thresholds to guide therapy decisions. Through the simultaneous use of risk equations and previously set thresholds, an optimal and cost-effective management plan can be provided to the patient. This is currently exemplified by an approach using the KFREs with risk thresholds to prompt transition from one stage of care to the next (Fig. 19.7).[66,335]

PRIMARY CARE TO NEPHROLOGY TRANSITION

In the initial stages of CKD, patients often develop salt and water retention, hypertension, and anemia.[336] Primary care physicians are able to manage these symptoms and provide optimal care for these patients, avoiding both unnecessary spending and treatments. However, when patients exhibit a high risk of CKD progression, referral to a nephrology service and subsequent access to interdisciplinary care teams is necessary to delay CKD progression, prevent suboptimal dialysis initiation, and reduce the chances of early mortality.[7,337] The use of risk equations and risk thresholds may allow physicians to be better informed about when such a referral should be considered. This approach may also prevent low-risk CKD patients from exposure to unnecessary treatments and the anxiety associated with referral.[338] In 2021 the National Institute for Health and Care Excellence guideline for the diagnosis and management of CKD recommended a four-variable KFRE predicted risk of kidney replacement therapy >5% at 5 years as a criterion for referral from primary care to nephrology, and in 2024 KDIGO recommended a similar approach using a 5-year predicted risk threshold of 3% to 5%.[339,340] Integrating risk

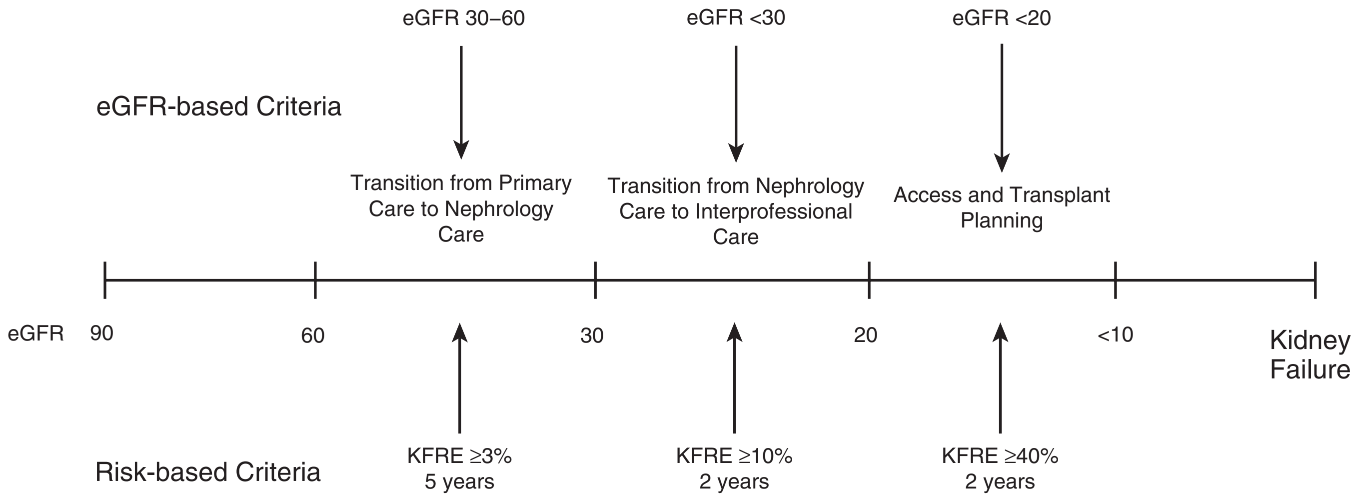

Fig. 19.7 A risk-based versus estimated glomerular filtration rate (eGFR)-based approach to clinical decision making in patients with chronic kidney disease (CKD). *KFRE,* Kidney failure risk equation. (Reproduced with permission from Tangri N, Ferguson T, Komenda P. Pro: risk scores for chronic kidney disease progression are robust, powerful and ready for implementation. *Nephrol Dial Transplant*. 2017;32:748–751.)

prediction models into clinical care can also be potentially useful in the planning of follow-up. Patients with a high risk of CKD progression but only moderately reduced eGFR can be followed at 3- or 6-month intervals, whereas those with lower risk can be assessed annually.[341]

INTERDISCIPLINARY CARE

The use of an interdisciplinary nephrology team, which involves a nurse, pharmacist, and dietician, may improve concordance with treatment, reduce suboptimal dialysis initiation, and could even reduce mortality for high-risk CKD patients. As with most specialized clinical services, inclusion of too many patients with a low risk of CKD progression can saturate interdisciplinary resources with minimal benefits.[335] This was a major issue in Ontario, Canada, where health care administrators struggled with excess demand for interdisciplinary care with a limited health care budget. As a result, in 2016 the Ontario Renal Network, which is responsible for administering the provinces' renal health resources, adopted a risk-based cutoff for eligibility of clinical resources. Patients with a KFRE-calculated 2-year risk of ESRD exceeding 10% or an eGFR <15 mL/min/1.73 m^2 were eligible for interdisciplinary care. This drastically lowered the number of patients eligible for interdisciplinary care as the previous threshold for such treatment was an eGFR <30 mL/min/1.73 m^2, irrespective of risk. It was determined using the KFREs that only two-thirds of patients who fit the previous criteria had a 2-year risk >10%. Also, from implementation of these new criteria, a small proportion of patients with an eGFR between 30 and 45 mL/min/1.73 m^2 were determined at high risk of CKD progression and were now eligible for interdisciplinary care. This reallocation of resources led to cost savings of approximately 6 million Canadian dollars annually and allowed high-risk patients with an eGFR >30 mL/min/1.73 m^2 earlier access to interdisciplinary care.[342]

DIALYSIS ACCESS AND MODALITY PLANNING

Current clinical practice guidelines recommend dialysis modality planning for patients with an eGFR <30 mL/min/1.73 m^2. However, the majority of these patients, particularly the elderly, may never actually progress to ESKD.[343] For these patients, the prospect of dialysis creates unnecessary anxiety and results in unnecessary treatment planning. In particular, the risk-benefit balance for arteriovenous fistula (AVF) insertion is important. While AVF is the standard form of dialysis access for patients on hemodialysis, the benefits may be reduced in the elderly. These individuals suffer from a higher likelihood of primary AVF failure and adverse events including heart failure and steal syndrome. Older persons at low risk of developing ESRD also tend to have a higher risk of competing morbidities, thereby making preemptive AVF formation both futile and costly.[344] The use of a risk prediction model to guide the timing of AVF formation can overcome some of these limitations and provide more optimal treatment plans. Patients with low-risk CKD progression will adopt a monitoring-based approach, whereas those with the highest risk of progression can be recommended for access placement for dialysis. In 2014, a Markov model was developed for assessing the implementation of the KFREs on simulated CKD progression in the Sunnybrook hospital cohort. The goal was to determine if the KFREs could select, with the highest accuracy, those patents in need of access for dialysis, while simultaneously avoiding creation of access in patents who either died before ESRD or who did not progress to dialysis within 2 years of access insertion. It was found the KFRE-based threshold of 20% to 40% risk was superior to eGFR-based thresholds in predicting the need for dialysis and AVF formation.[345]

MODELS TO PREDICT CHRONIC KIDNEY DISEASE PROGRESSION IN EARLIER STAGES OF DISEASE

Newer therapies to slow CKD progression such as SGLT2 inhibitors and nonsteroidal MRAs have changed the landscape of CKD management, and high-risk patients, if treated early with multidrug therapy can prevent kidney failure in later life. Identification of patients at high risk of CKD progression before the majority of kidney function

Table 19.6 Overview of Models that Predict a 40% Decline in eGFR or Dialysis

	CKD-PC[308]	Klinrisk[307]	KidneyIntelX[241]
Variables	16 variables including sBP, diabetes or antihypertensive medication use, history of select heart diseases, and smoking history	22 laboratory variables derived from CBC, chemistry panel, and urine sample	3 proprietary biomarkers, 7 additional clinical variables including albuminuria, sBP, and SCa
Outcome	3-year probability of 40% decline in eGFR	2- and 5-year probability of 40% decline in eGFR	Composite outcome of eGFR decline of ≥5 mL/min per year, ≥40% sustained decline, or kidney failure within 5 years
Total population	1.6 million adults, with or at risk for CKD	177,196 adults with CKD G1-G4 or at risk for CKD	1146 adults with CKD (G1-G3) and diabetes
Cohort demographics	43 cohorts globally	From Manitoba and Alberta Canada, U.S. Payers, FIDELITY clinical trials, CANVAS CREDENCE clinical trials	From Mount Sinai and Penn Hospital Systems, CANVAS CREDENCE clinical trials
Time horizon	5 years	1-5 years	5 years
Discrimination and calibration	0.74-0.77[a]	0.84-0.88[a]	0.77[a]

[a]AUC scores refer to initial development and external validation.
CBC, Complete blood count; *CKD,* chronic kidney disease; *CKD-PC,* Chronic Kidney Disease Prognosis Consortium; *sBP,* systolic blood pressure; *SCa,* serum calcium.

is lost could help to attain the maximal benefit from introduction of disease-modifying therapy.[346-348]

Some studies have examined the performance of the KFRE in earlier stages of disease (eGFR > 60 mL/min) and found that the equation does not discriminate or calibrate well. As such, newer equations using both regression and machine learning approaches to predict a more proximal endpoint (40% decline in eGFR or dialysis) have been developed and are now externally validated (Table 19.6).[307,241,308]

Integration of these models into electronic health records or laboratory information systems is feasible and can aid in early identification of patients with high-risk CKD, as well as help patients, payers, and health systems achieve the maximum possible benefit from multidrug treatment.[349,350]

FUTURE CONSIDERATIONS

Considerable progress has been made in identifying risk factors that predict the progression of CKD in diverse cohorts from the general population and nephrology clinics. There is some variation among studies, likely due to differences in the populations and variables studied. Remarkably, a relatively small group of risk factors appears common to many studies, and much progress has been made in developing risk scores based on these variables to predict CKD progression.

Future studies will likely focus on the use of novel biomarkers and genetic factors as risk factors and variables in risk scores, although measurement of such markers is likely to be associated with greater cost than the simple risk factors used to date. Accurate risk scores can align risk and resources while allowing for the most personalized care for persons with CKD. As a result, now is the time to implement these risk scores into clinical care to help patients and clinicians navigate the transition from CKD to dialysis. Further studies are also required to develop risk scores to predict cardiovascular risk in patients with CKD that account for the close association between CKD and CVD.

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QUESTIONS

1. Which of the following is FALSE regarding risk factors for chronic kidney disease (CKD) progression?
   a. More rapid declines in estimated glomerular filtration rate (eGFR) are common in younger patients with CKD and lead to a higher risk of kidney failure and death.
   b. The etiology of CKD is not relevant when considering the risk of progression to ESKD.
   c. Ethnicity is a strong indicator of CKD progression to end-stage kidney disease (ESKD), and African Americans are at higher risk.
   d. Coding variants in the *APOL1* gene are associated with an increased risk for development of ESKD.

   Answer: b

   Rationale: Some causes of CKD are associated with consistently higher risk of CKD progression than others. For example, adult polycystic kidney disease is associated with the highest risk of ESKD. Diabetic kidney disease also tends to be associated with a higher risk of CKD progression than many other causes.

2. Which of the following biomarkers may NOT be useful in risk stratification of patients with chronic kidney disease (CKD)?
   a. Asymmetric dimethylarginine (ADMA)
   b. Neutrophil gelatinase–associated lipocalin (NGAL)
   c. Alpha-fetoprotein (AFP)
   d. Uromodulin (UMOD)

   Answer: c

   Rationale: ADMA, NGAL, and UMOD have all been shown to be useful biomarkers in the risk stratification of patients with CKD.

3. Which of the following is accurate regarding the relationship between cardiovascular disease and chronic kidney disease (CKD)?
   a. There is only a weak association between CKD progression and death due to cardiovascular disease.
   b. Diuretics are contraindicated in treatment of sodium retention–induced hypertension for patients with CKD.
   c. Current cardiovascular risk prediction models require further validation in persons with CKD, particularly for those with advanced CKD (stages 4–5).
   d. All of the above

   Answer: c

   Rationale: Current risk scores for CVD have shown poor prediction model metrics in patients with CKD.

4. The kidney failure risk equations (KFREs) use which of the following variables when predicting the development of end-stage kidney disease (ESKD) in patients with chronic kidney disease (CKD)?
   a. Estimated glomerular filtration rate (eGFR)
   b. Albuminuria
   c. Age
   d. Sex
   e. All of the above

   Answer: e

   Rationale: The KFRE uses eGFR + Albuminuria + Age + Sex, to predict ESKD in patients with CKD.